

DSCE - Tilburg University

**Robust search policy design for drone swarms in
adversarial hide & seek environment**

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Github repository link: <https://github.com/J4nvg/Internship>

1 Introduction

Discrete search games were first established in (Bram, 1963), presenting a scenario where a hider selects one of n boxes for a target, and a searcher opens them sequentially to minimize the expected number of inspections

(Lidbetter, 2025). This framework was expanded by (Lidbetter, 2020), who introduced the concept of a "success probability." This addition models a dangerous environment where the searcher is not guaranteed to survive the expansion of a box, finding a theoretical solution for a single searching agent.

While drone swarms are a rapidly emerging technology that offers opportunities for search and rescue operations (Hoang et al., 2023; Arnold et al., 2020; Alfeo et al., 2018), existing research has not integrated this danger component. In high-stakes scenarios—such as navigating hazardous terrain or operating in adversarial environments with a risk of interception, accounting for agent loss is critical. Therefore, basing the approach on the single-searcher literature by (Lidbetter, 2020) has the highest potential of succeeding in finding a heuristic or even exact solution.

2 Environment and Definitions

Hiders and Strategies The game features n static hiders placed in a subset of candidate locations S . Candidate cells are selected uniformly at random, and hiding probabilities q_i are assigned based on three strategies: (1) **Greedy**: Hiders select the subset of cells with the highest q_a values (derived from (Lidbetter, 2020) Eq. 1); (2) **Weighted**: Hiders select cells probabilistically based on the distribution q_a (proven optimal for single-searchers); (3) **Random**: Probabilities q_i are drawn from a Dirichlet distribution such that $\sum q_i = 1$.

Success Probability (Risk) Each location $i \in S$ has a *success probability* p_i , representing the chance a searcher survives inspecting that cell. three risk environments were modeled:

- **High Variance**: $p_i \in \{0.10, 0.30, 0.60, 0.80, 0.95\}$;
- **Low Variance**: $p_i \in \{0.60, 0.62, 0.64, 0.66, 0.68\}$;
- **Skewed**: $p_i \in \{0.60, 0.62, 0.64, 0.66, 0.10\}$;

Swarm Dynamics The swarm enters the grid at (0,0) and operates under the following constraints: (1) Movement is restricted to cardinal directions (no diagonals); (2) Drones know candidate locations and success probabilities (p_i) but *not* the hiding probabilities (q_i) but tactics can of course re-calculate this; (3) Discovery is deterministic: if a drone expands a hider’s cell and survives, the hider is found ($P(\text{Found}|\text{Hider present} \wedge \text{Survival}) = 1$). Conversely, if the drone is taken down, the hider remains undiscovered.

3 Searching policies

3.1 Game-Theoretic & Risk-Aware

3.1.1 Divide over risk - Proportional Risk split

This strategy allocates the drone swarm to candidate locations proportionally to the risk associated with each location, defined as ‘1 - probability of success’. The goal is to send the largest groups of drones to the highest-risk locations first.

First, all candidate locations are sorted by risk. Then, a fractional number of drones is calculated for each location based on its proportional risk (e.g., 7.44 drones for location A, 2.56 for B). The algorithm assigns the *floor* of this number to each location (7 to A, 2 to B). The remaining drones (the "remainder") are then assigned one by one to the locations with the largest fractional parts (B would get the first remainder, as $0.56 > 0.44$).

In the routing phase, all drones are assigned to visit all candidate locations. Each drone group’s route is a `visit_order` list that is rotated to start with that group’s primary assigned target. For example, if the ordered risk locations are [A, B, C]:

- Group A (highest risk): [A, B, C]
- Group B (mid risk): [B, C, A]
- Group C (low risk): [C, A, B]

3.1.2 Lidbetter

In this searching strategy the cells are explored based on the optimal searching strategy given by (Lidbetter, 2020) . A subset of possible hiding locations is chosen with probability q_A , after searching those, the remaining possible hiding locations are chosen uniformly at random. The swarm traverses all cells together. ($A \in S^{(k)}$) with probability

$$q_A = \lambda_k \prod_{i \in A} \frac{1 - p_i}{p_i}, \text{ where } \lambda_k = \left(\sum_{B \in S^{(k)}} \prod_{i \in B} \frac{1 - p_i}{p_i} \right)^{-1}. \quad (1)$$

Contrary to other strategies the Lidbetter strategy was implemented to specifically try to find a route that does not come across any of the possible hiding locations that are not the main target.

3.1.3 Traverse ordered qa, Ranked qa subset

This strategy assigns each drone to a different subset $A \in S^{(k)}$ based on the q_A values from Equation 1. All subsets are sorted in descending order by q_A , creating an ordered list where index 0 corresponds to the highest-value subset.

Drones are assigned sequentially: drone 1 receives the subset at index 0 (highest q_A), drone 2 receives index 1, and so on. If the number of drones exceeds the number of subsets, the assignment wraps around. For example, with 4 drones and 3 subsets, drone 4 would be assigned the subset at index 0 again.

After receiving its primary subset, each drone visits those cells first, then visits all remaining candidate locations in order of the list, like the primary target assignment wrapping around if the list gets exhausted.

3.1.4 Traverse P qa, Weighted qa subset sampling

This strategy independently samples a subset $A \in S^{(k)}$ for each drone according to the distribution q_A from Equation 1. This then is its first objective, where after it then traverses all remaining candidate cells at uniform random order. Such that its complete route consists of all possible hiding locations. Whereas the Lidbetter policy in 3.1.2 has the swarm collectively choose a single subset according to q_A and then visit the remainder uniformly at random together, this variant applies the same distribution per drone.

3.2 Communication enabled

3.2.1 Shared list, Dynamic task queue

Different from the other policies that are given an initial route and then fly it, this searching strategy dynamically assigns new locations. I.e. allowing the drones to communicate during the mission. Let S be the set of possible hiding locations. Initially, up to $\min(\text{swarm_size}, |S|)$ drones are assigned to hiding locations from S , leaving the remaining drones idle. The active and idle states are tracked, along with a 'to search' queue and a set of successfully searched locations. When a drone is assigned a location, that location is removed from the 'to search' queue.

If a drone dies either during route traversal or while searching the location, that location is appended back to the 'to search' queue. The closest idle agent (measured by Manhattan distance) is then assigned this target. The route given to this reassigned agent avoids successfully searched locations.

When a drone successfully completes its route and searches a location, that location is added to the successfully searched set, and the drone returns to the idle state where it can be reassigned to remaining locations in the queue.

Note that there is no priority given to cells with a higher $1 - p_i$ such that these are searched first. This could be a future improvement such that smaller sized swarms are more successful.

3.3 Cost-Optimization Heuristics

3.3.1 Together traverse best permutation, Shortest path tour

This strategy calculates the optimal permutation of cells to visit, such that the total distance traveled is minimized. Given that the environment is a grid that does not allow diagonal movement, the manhattan distance formula is used to calculate the distance: $|x_1 - x_2| + |y_1 - y_2|$. The simulated implementation brute

forces all options, and does not use any of the Traveling Salesman Problem optimizations. The swarm traverses all cells together.

3.3.2 Discounted Distance - Risk weighted tour

This strategy selects the permutation of candidate cells that minimizes the total travel costs over the entire route, i.e., it takes the permutation whose total cost is minimal under the cost function below. Given consecutive cells $p_1 = (x_1, y_1)$ and $p_2 = (x_2, y_2)$, the travel cost is

$$cost(p_1, p_2) = (|x_1 - x_2| + |y_1 - y_2|) \times (1 - P_{p_2}) \quad (2)$$

where the distance term is the Manhattan metric and P_{p_2} is the success probability at the next cell; the total route cost is the sum of these costs, and the implementation brute-forces all permutations to find the minimum-sum one with the swarm traversing together.

3.3.3 Discounted Distance Reverse - Success weighted tour

This policy is almost identical to 3.3.2 Yet now the discount factor is P_{p_2} , yielding the cost function:

$$cost(p_1, p_2) = (|x_1 - x_2| + |y_1 - y_2|) \times P_{p_2} \quad (3)$$

The simulated implementation brute forces all options, and does not use any of the Traveling Salesman Problem optimizations. The swarm traverses all cells together.

3.4 Geometric, Pattern based

3.4.1 Horizontal scan traversal - Horizontal scan

In this searching strategy drones collectively traverses the complete grid starting from Top-left and ending on the bottom-left. This is done in the order: left to right, down, right to left, down, repeat. Thus making a 'snake-like' movement pattern.

3.4.2 Spiral scan traversal - Inward Spiral

In this searching strategy drones collectively traverse the grid in a spiraling pattern. Moving together from top left to bottom left, to bottom right, top right, back to the top left and going inwards right before the first cell. Continuing this movement until all cells have been expanded.

3.4.3 Partitioned horizontal scan traversal - Partitioned horizontal scan

This searching strategy is similar to the horizontal scan in 3.4.1, yet drones now independently traverse the grid from Top-left to bottom-left. They too start at the top left, but continue to divide themselves (as evenly as possible) over the vertical axis. Effectively scanning the board in partitions, making the same snake like pattern, but starting the movement at different parts of the board.

3.5 Stochastic

3.5.1 Random Walk

In this search policy every drone can, at each time step t , independently choose to move in one of the North, East, South, West directions in uniform random order, restricted by the bounds of the grid.

4 Method

4.0.1 Simulation Environment

A custom discrete-event simulator was developed using Python to evaluate the performance of drone swarm search policies. The simulator utilizes an object-oriented architecture to ensure modularity and code readability, allowing for distinct variations in search policies, hiding strategies, and environmental parameters. The simulation models an $N \times N$ grid world where movement is restricted to cardinal directions (North, East, South, West). To account for the stochastic nature of detection and the initialization of hider locations, a Monte Carlo simulation approach was adopted.

4.0.2 Experimental Design

The experiments were designed to test the performance of search policies under a wide variety of hiding conditions, employing the independent variables detailed in Table 1. A total of 100,000 simulation trials were conducted to yield the dependent variables. This sample size ensures statistical significance, resulting in a 95% confidence interval margin of error of $\epsilon \approx 0.005$ for the primary success metric, $P(\text{all found})$.

Table 1: Summary of simulation parameters and game settings.

Variable	Value(s)
Grid size	20×20
Swarm sizes	$\{1, 5, 10\}$
Possible hiding spots	5
Hider count	$\{1, 2, 3, 4, 5\}$
Hiding strategies	$\{\text{Weighted, Random, Greedy}\}$
Success probability distributions	$\{\text{Low_Var, High_Var, Skewed}\}$
Trials	100,000

For every simulation configuration, consisting of the game settings in Table 1 paired with a specific search policy, the metrics listed in Table 2 were recorded to quantify performance.

Table 2: Dependent variables (metrics) recorded during simulation.

Metric	Description
$P(\text{all found})$	The probability that all hiders are found (measure of success).
Steps	The total number of simulation steps until termination (measure of time).
Mean distance traveled	The average Manhattan distance traveled by all drones in the swarm until termination.
Fraction of hiders found	The proportion of total hiders located before the game terminated.
Drones taken down	The number of drones captured/disabled before termination.

4.0.3 Performance Evaluation

To determine the optimal policy, specific criteria were established. A search policy is credited with a "win" if it is the highest-performing tactic in a specific scenario. In cases where two or more policies perform equally well—defined as sharing the same $P(\text{all found})$ when rounded to three decimal places—all top-performing policies receive a win.

However, relying solely on win counts is insufficient, as the scenarios require a more nuanced evaluation. A tactic may excel in one scenario yet fail in another. Therefore, additional performance measures are introduced to identify a policy that is robust across both different environments and hiding tactics. These measures are not only intuitive but also commonly used, such as the measure of spread in human-robot teaming systems (Rothfus, Chiou, Inga, Hohmann, & Stolkin, 2022; Nielsen, Few, & Athey, 2008) and maximin criterion maximizing the probability of success in the worst case, also aligning with the goal mentioned in (Lidbetter, 2020), .

Let E denote the set of environmental values, as displayed in Table 1, and let Π denote the set of searching policies as discussed in Section 3.

1. **Performance Spread (Robustness):** The difference between the best and worst performance. A *smaller* spread indicates higher stability and robustness.

$$\Delta P = \max(P(\text{All Found}|\pi, e)) - \min(P(\text{All Found}|\pi, e)) \quad (4)$$

2. **Maximin Criterion:** This criterion selects the policy that maximizes the minimum success probability

across all variations.

$$\pi_{\maximin}^* = \operatorname{argmax}_{\pi \in \Pi} \left(\min_{e \in E} P(\text{All Found} | \pi, e) \right) \quad (5)$$

5 Results

Figure 1 illustrates the win counts for the various policies. In single-drone settings, the *Lidbetter* policy achieved the highest number of wins (8). In five-drone swarms, the *Proportional risk split* tactic received the most wins (17), while the *Dynamic task queue* policy performed best in ten-drone swarms, achieving the highest overall performance count (24).

Based on these win counts, an initial conclusion regarding the optimal policy per swarm size can be drawn. However, the robustness analysis defined in the previous section yields further insights.

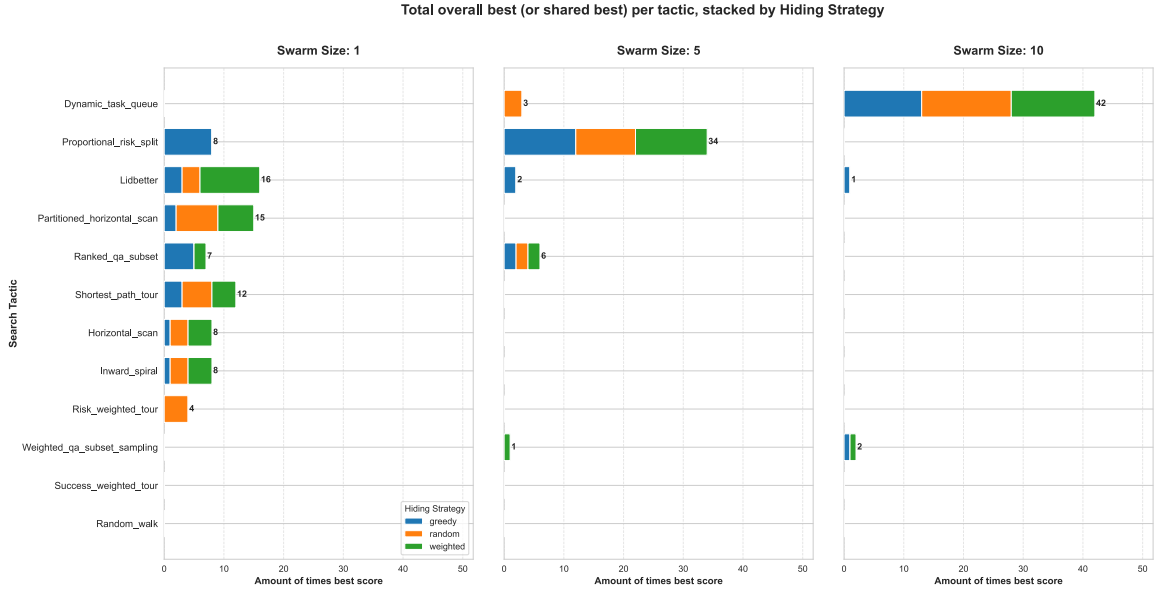


Figure 1: Stacked bar plot of the amount of times a search policy had the highest P(all found) in all different environment conditions. Blue are the contributions from the greedy hiding strategy, orange for the random hiding strategy and green for the weighted hiding strategy.

5.1 One Hider

Prob. Dist.	Metric	Swarm size 1	Swarm size 5	Swarm size 10
Low var	$\pi_{\maximin}^*$	Lidbetter: 0.315	Ranked Q_a subset & proportional risk split: 0.852	Dynamic Task Queue: 0.985
	$\pi_{\max}$	Proportional risk split: 0.562	Ranked Q_a subset: 0.864	Dynamic Task Queue: 0.988
High var	$\pi_{\maximin}^*$	Lidbetter: 0.079	Weighted Q_a subset sampling: 0.322	Weighted Q_a subset sampling: 0.542
	$\pi_{\max}$	Risk Weighted Tour: 0.241	Dynamic Task Queue: 0.632	Dynamic Task Queue: 0.822
Skewed	$\pi_{\maximin}^*$	Lidbetter: 0.086	Weighted Q_a subset sampling: 0.344	Weighted Q_a subset sampling: 0.567
	$\pi_{\max}$	Risk Weighted Tour: 0.209	Ranked Q_a subset: 0.648	Dynamic Task Queue: 0.845

Table 3: Tabular view of the tactic $\pi_{\maximin}^*$ and $\pi_{\max}$ and the corresponding $P(\text{all found})$ value for 1 hider, hider ratio: $\frac{1}{5}$

swarm size 1 When analyzing the robustness of the searching policies across different probability distributions for a swarm size of 1 as show in 2, the majority of search tactics show a comparable and relatively narrow performance spread. π_{maximin}^* is the *Lidbetter* policy with the highest minimum success probability (0.079) , also aligning with the findings in table 3. As for the π_{max} in the comparison against all probability distributions in figure 2, two tactics - *ranked_qa_subset* and *Proportional_risk_split* - show a clear spike in performance for the greedy hiding strategy. This performance spike can be easily explained by the underlying mechanics of both algorithms. Although the algorithms do not learn or adapt and thus cannot overfit in the classical AI sense, their deterministic selection criteria are highly exploitative in these context. *ranked_qa_subset* prioritizes the cell with the highest q_a subset, while *Proportional_risk_split* targets the lowest p_i , which in turn effectively also targets the highest q_a values, as the q_a values are determined by the relation of $1 - p_i$. The prioritization of the highest q_a subset effectively counters the greedy hiding algorithm, which operates on a similar maximization logic. From table 3 it can be derived that *Risk_weighted_tour* shows up twice as π_{max} , this is can be contributed to the relatively high performance against the 'random' hiding strategies as can be seen in figures 7 , 9, 11.

swarm size 5 and 10 For the situation where the swarm size is 5, 2 shows that $\pi_{\text{maximin}}^* = \text{Weighted_qa_subset_sampling}$ (0.322). Indicating that this is the safest searching policy to pick for this swarm size. It can also be derived from the same figure that for swarm size 10, again, $\pi_{\text{maximin}}^* = \text{Weighted_qa_subset_sampling}$ with a success probability of (0.542). For swarm size 5 and swarm size 10 , respectively, $\pi_{\text{max}} = \text{ranked_qa_subset}$ with success probability 0.864 and $\pi_{\text{max}} = \text{dynamic_task_queue}$ with success probability 0.988. These findings align with the expectation set by the stacked bar plot in figure 1.

Therefore, in a context of a single hider, or even more specifically in this case where the ratio between hiders and potential hiding locations is 1/5, it is the most safe - the option with highest success probability in the worse case scenario - option is to select the *Lidbetter* searching policy for a **single searcher**, and the *Weighted qa subset sampling* searching policy if there are **5 or 10 searchers** in a swarm.

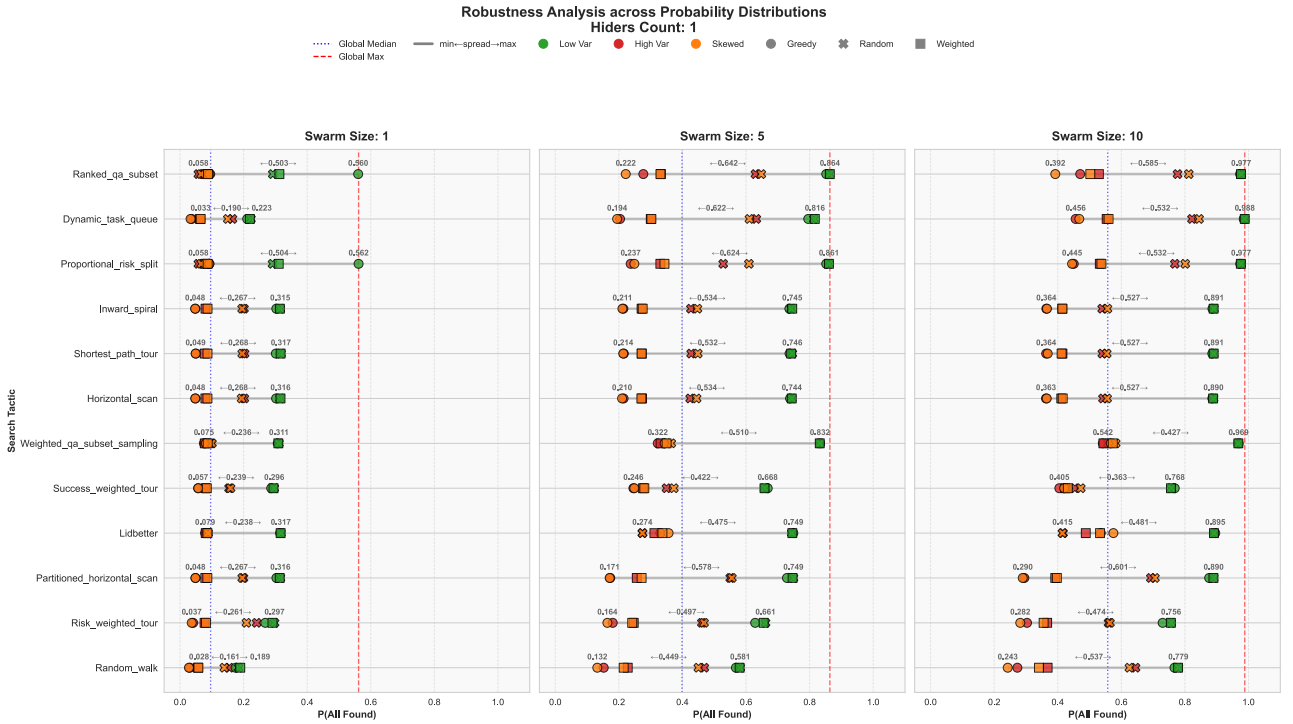


Figure 2: Cleveland dot plot of probability of all hiders found for swarm size 1, 5 and 10, given 1 hiders, 5 possible hiding locations, simulated on a 20 x 20 grid. The plot shows robustness of strategies across the different probability distributions and different hiding strategies, i.e. wider lines could be viewed as less robust.

5.2 Two Hiders

Prob. Dist.	Metric	Swarm size 1	Swarm size 5	Swarm size 10
Low var	$\pi_{\maximin}^*$	Lidbetter: 0.182	Proportional risk split: 0.760	Dynamic Task Queue: 0.976
	$\pi_{\max}$	Proportional risk split: 0.345	Proportional risk split: 0.773	Dynamic Task Queue: 0.979
High var	$\pi_{\maximin}^*$	Lidbetter: 0.025	Proportional risk split: 0.161	Dynamic Task Queue: 0.395
	$\pi_{\max}$	Risk Weighted Tour: 0.084	Dynamic Task Queue: 0.401	Dynamic Task Queue: 0.681
Skewed	$\pi_{\maximin}^*$	Lidbetter: 0.017	Proportional Risk split: 0.204	Dynamic Task Queue: 0.445
	$\pi_{\max}$	Risk Weighted Tour: 0.077	Proportional Risk split: 0.407	Dynamic Task Queue: 0.719

Table 4: Tabular view of the tactic $\pi_{\maximin}^*$ and $\pi_{\max}$ and the corresponding $P(\text{all found})$ value for 2 hiders, hider ratio : $\frac{2}{5}$

swarm size 1

When analyzing the robustness of the searching policies across different probability distributions for a swarm size of 1, as shown in Table 4 and 3, the Lidbetter policy consistently proves to be the most robust strategy. $\pi_{\maximin}^*$ is the Lidbetter policy across all distributions, achieving the highest minimum success probability in the Low Var (0.182), High Var (0.025), and Skewed (0.017) scenarios. Regarding $\pi_{\max}$ in the comparison against all probability distributions, the ranked q_a subset and Proportional Risk Split tactics show a distinct spike in performance. As discussed in section 5.1, this is due to their underlying mechanics exploiting the greedy hiding strategy.

Swarm size 5 and 10

For the situation where the swarm size increases to 5, Table 4 shows a shift in the optimal policy. Here, $\pi_{\maximin}^* = \text{Proportional Risk Split}$, with success probabilities of 0.760 (Low Var), 0.161 (High Var), and 0.204 (Skewed). This indicates that for a medium-sized swarm relative to the environment, Proportional Risk Split is the safest searching policy to pick. It is also worth noting that in the Skewed distribution, this policy serves as both the maximin and the absolute maximum choice.

For swarm size 10, the Dynamic Task Queue becomes the dominant strategy. It achieves $\pi_{\maximin}^*$ across all distributions—Low Var (0.976), High Var (0.395), and Skewed (0.445)—while simultaneously holding the $\pi_{\max}$ in all three scenarios. This finding strongly aligns with the win counts presented in 1, where Dynamic Task Queue achieved the highest overall performance count for 10-drone swarms.

Therefore, in a context of two hiders where the ratio between hiders and potential hiding locations is $2/5$, the safest option with the highest success probability in the worst-case scenario is to select the Lidbetter policy for a single searcher. As the swarm size increases, the recommendation shifts: Proportional Risk Split is optimal for 5 drones, while the Dynamic Task Queue is the robust and performant choice for a swarm of 10 searchers.

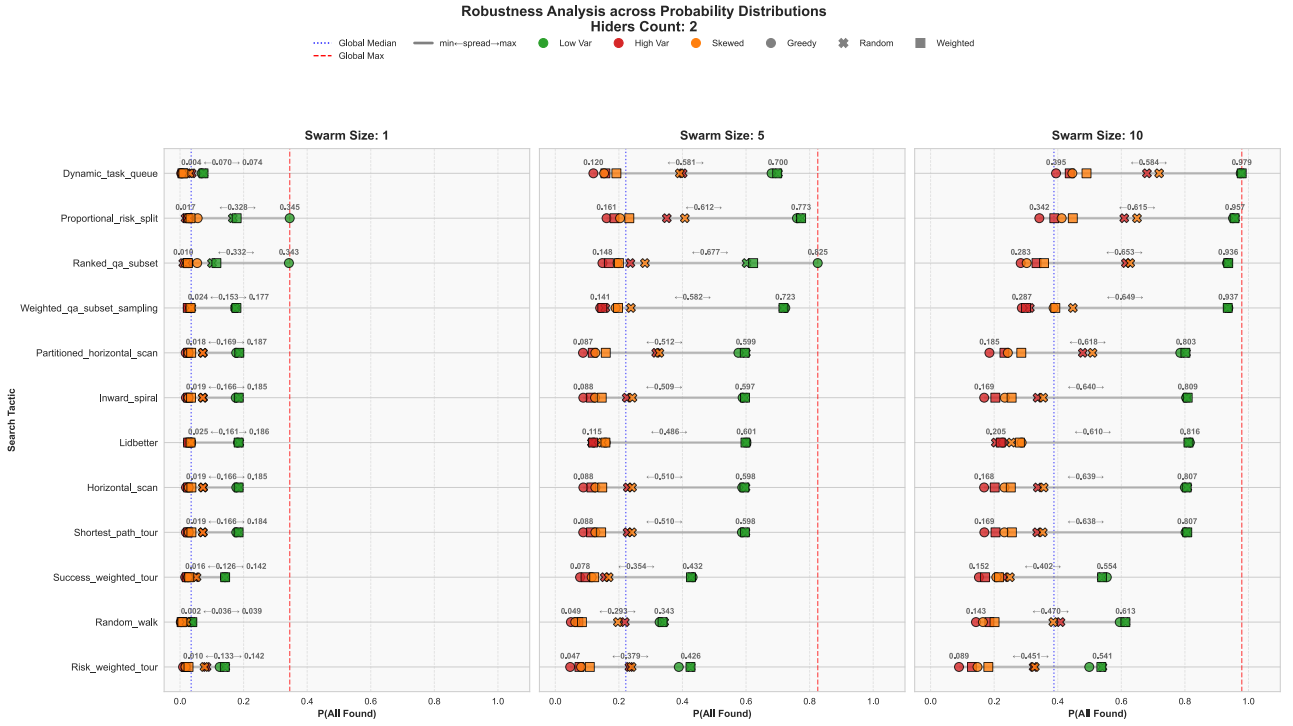


Figure 3: Cleveland dot plot of probability of all hidrs found for swarm size 1, 5 and 10, given 2 hidrs, 5 possible hiding locations, simulated on a 20 x 20 grid. The plot shows robustness of strategies across the different probability distributions and different hiding strategies, i.e. wider lines could be viewed as less robust.

5.3 Five Hiders

Prob. Dist.	Metric	Swarm size 1	Swarm size 5	Swarm size 10
Low var	$\pi_{\max}^*, \pi_{\max}$	Lidbetter, shortest path tour, (Partitioned) horizontal scan: 0.106	Proportional risk split: 0.625	Dynamic Task Queue: 0.961
High var	$\pi_{\max}^*, \pi_{\max}$	Lidbetter, shortest path tour, (Partitioned) horizontal scan, inward spiral: 0.013	Proportional risk split: 0.120	Dynamic Task Queue: 0.389
Skewed	$\pi_{\max}^*, \pi_{\max}$	Lidbetter, shortest path tour, (Partitioned) horizontal scan, inward spiral: 0.015	Proportional Risk split: 0.142	Dynamic Task Queue: 0.430

Table 5: Tabular view of the tactic $\pi_{\max}^*$ and $\pi_{\max}$ and the corresponding $P(\text{all found})$ value for 2 hidrs, hider ratio : $\frac{2}{5}$

It must be noted that for the five hidrs case, the amount of hidrs in this scenario is saturated (5/5). Effectively rendering the hiding strategies redundant as all possible hiding locations will be filled anyways now. Therefore in this special case, when looking at the performance given a said probability distribution $\pi_{\max}^* = \pi_{\max}$.

swarm size 1

When analyzing the robustness of the searching policies across different probability distributions for a swarm size of 1 with 5 hidrs as shown in 4, the majority of search tactics show a comparable and relatively narrow range between their minimum and maximum $P(\text{All Found})$ values. The geometric pattern-based policies mentioned in 3.4, **shortest_path_tour** and **Lidbetter** outperform the other tactics, closely followed by the game-theoretic and risk-aware policies from 3.1.

Trivially, for a single drone the **partitioned_horizontal_scan** policy functions identical to the standard

`horizontal_scan1`, as there is only one agent to traverse the partition. Additionally, both `inward_spiral` and `horizontal_scan` theoretically exhaust the maximum number of cells. Since potential hiding spots are selected randomly and both policies initialize a route covering the complete grid, their success probability is determined by the product of success chances of all cells in the grid. This exhaustive coverage explains the performance of these algorithms.

Conversely, the other searching policies do not guarantee such systematic coverage; Due to the fact that most policies do not keep track of already searched cells and due to calculating routes dynamically from point A to point B, they may bypass cells or re-traverse already expanded space, leading to higher variance in performance.

Swarm size 5 and 10

For the best searching policy in the case of 5 searchers, based on Figure 4, $\pi_{\max}^* = \text{Proportional risk split}$; this is also the $\pi_{\max}$, a result which trivially follows from Table 5. Regarding the case of 10 searchers, Figure 4 shows that $\pi_{\max}^* = \text{Dynamic task queue}$, which similarly serves as the $\pi_{\max}$ as detailed in Table 5.

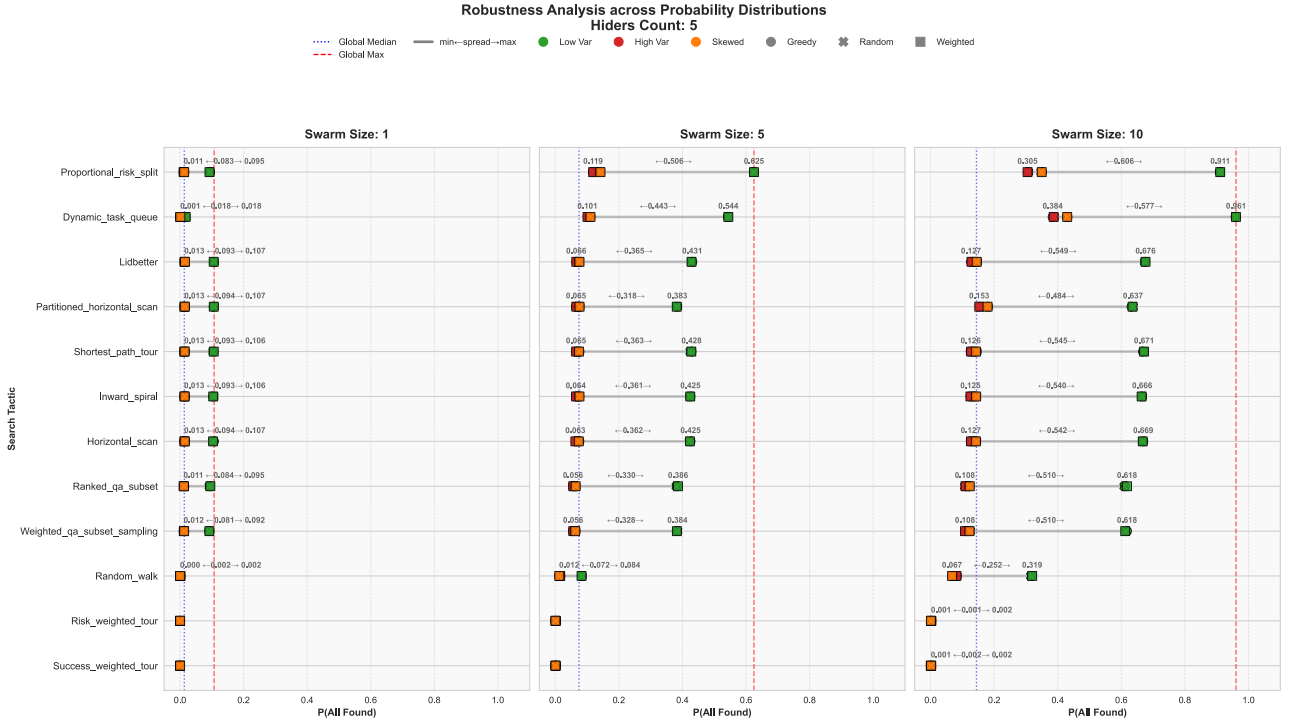


Figure 4: Cleveland dot plot of probability of all hidrs found for swarm size 1, 5 and 10, given 5 hidrs, 5 possible hiding locations, simulated on a 20 x 20 grid. The plot shows robustness of strategies across the different probability distributions and different hiding strategies, i.e. wider lines could be viewed as less robust.

5.4 Probability Distributions

In the case that a probability distribution about the environment is known, table 6 shows that the most robust tactic for swarm size 1 is Lidbetter among with other tactics. The result here is debatable as it the Lidbetter strategy is proven to be the most optimal searching policy, thus the implementation of the policy might have some flaws that make it less optimal.

swarm size 5 and 10 for 5 and 10 searchers the results align nicely with the previous discussed results. For 5 searchers the **proportional risk split** is the most robust strategy and for 10 searchers the Dynamic Task Queue is the most robust strategy.

This creates the believe that the probability distribution of the environment does not affect the searching policy by much.

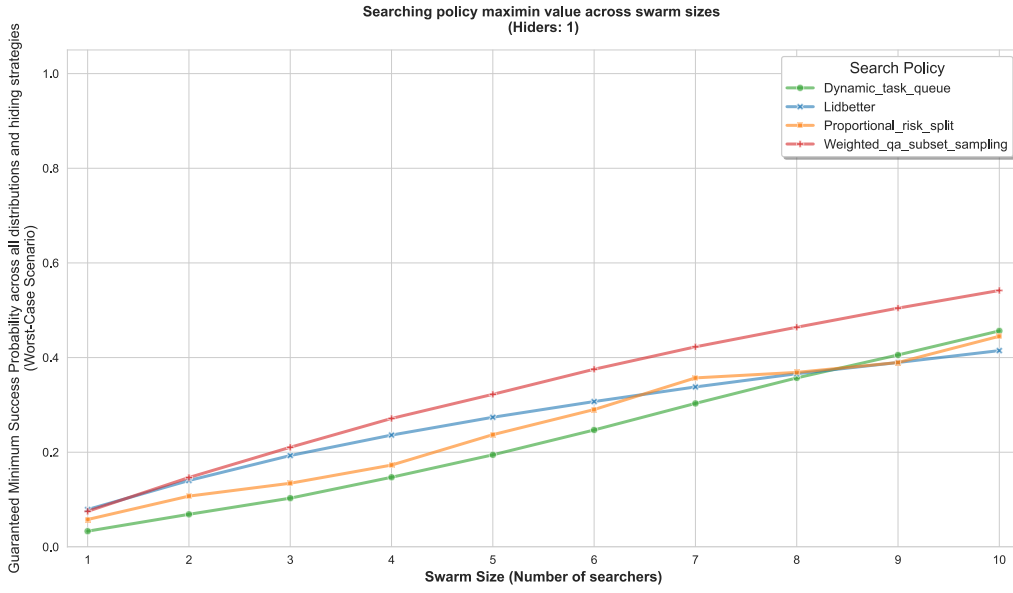
Prob. Dist.	Metric	Swarm size 1	Swarm size 5	Swarm size 10
Low var	$\pi_{\max}^*$	Lidbetter & (partitioned)horizontal scan: 0.106,	proportional risk split: 0.625	Dynamic Task Queue: 0.960
	$\pi_{\max}$	Proportional risk split: 0.562	Ranked Q_a subset: 0.864	Dynamic Task Queue: 0.988
High var	$\pi_{\max}^*$	Lidbetter,shortest path tour, (Partitioned) horizontal scan, inward spiral: 0.013	proportional risk split: 0.119	Dynamic Task Queue: 0.384
	$\pi_{\max}$	Risk Weighted Tour: 0.241	Dynamic Task Queue: 0.632	Dynamic Task Queue: 0.822
Skewed	$\pi_{\max}^*$	Lidbetter, shortest path tour,(Partitioned) horizontal scan, inward spiral: 0.015	proportional risk split: 0.140	Dynamic Task Queue: 0.429
	$\pi_{\max}$	Risk Weighted Tour: 0.209	Ranked Q_a subset: 0.648	Dynamic Task Queue: 0.845

Table 6: Tabular view of the tactic $\pi_{\max}^*$ and $\pi_{\max}$ and the corresponding $P(\text{all found})$ value for the different probability distributions. Showing all hider options.

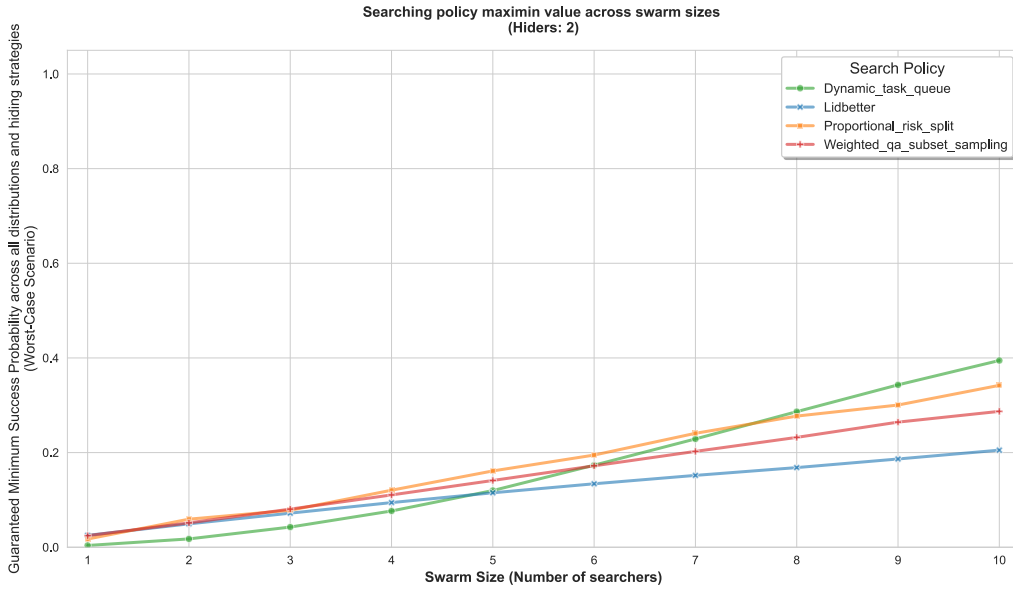
5.5 Across swarm sizes

Based on the previously discussed results there are four tactics that show to be promising, namely: **Lidbetter**, **Weighted_qa_subset_sampling**, **Proportional_risk_split**, **Dynamic_task_queue**. However each of these tactics show to be the most robust in different swarm sizes and hider ratios (actual hidens / the amount of possible hidens), as the *lidbetter* strategy often appears as most robust for swarm size 1 across all hider ratios, *proportional risk split* often appears as most robust for swarm size 5 across most, but not all, hider ratios, and *dynamic task queue* also appears to be the most robust for swarm size 10 across most but not all hider ratios. And lastly, *Weighted qa subset sampling* appears as most optimal across the 1 hider scenario for swarm sizes 5 and 10.

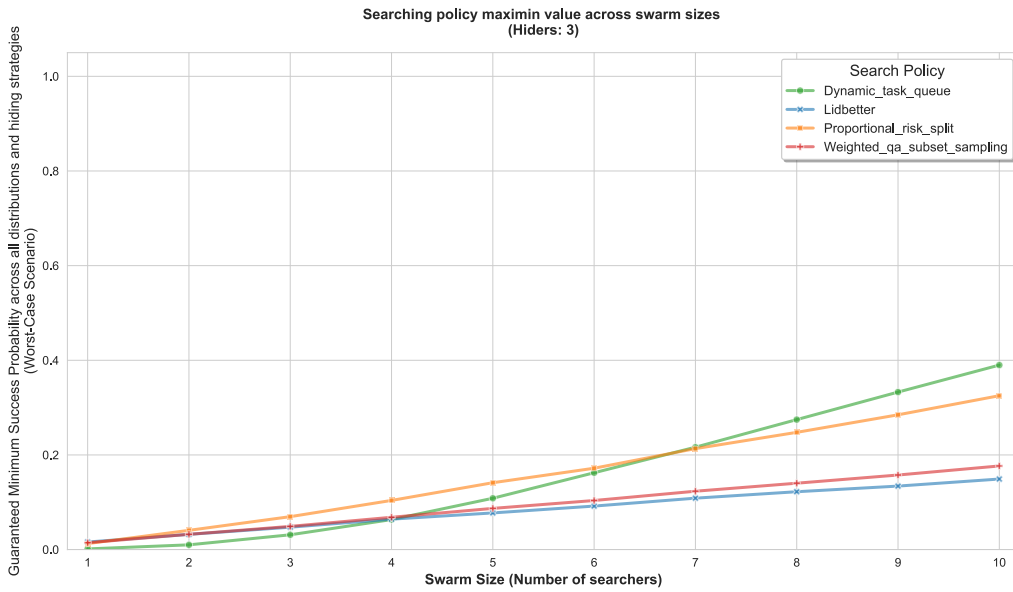
figure 5 show the turnarounds of when a tactic becomes the robust 'optimal' searching policy. For most hider ratios the optimal searcher strategy per amount of drones is roughly similar. Lidbetter performs best for 1 searcher (theoretical optimum), Proportional risk split from around 2, and then after a swarm size of 7 it becomes the dynamic task queue policy. Based on these findings a transition matrix can be built which then could function as a decision making tool. The transition matrix in figure 6 shows the most robust searching policy for a given drone swarm size and number of hidens. It is apparent from the figures that most of the optimal searching policies will remain mostly the same regardless of probability distributions.



(a) Hiders: 1

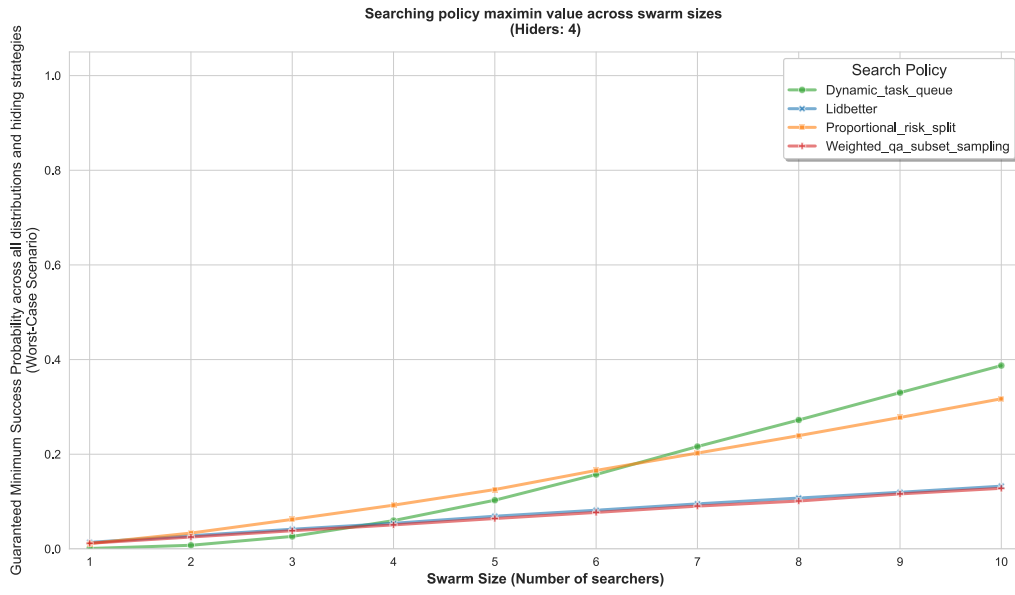


(b) Hiders: 2

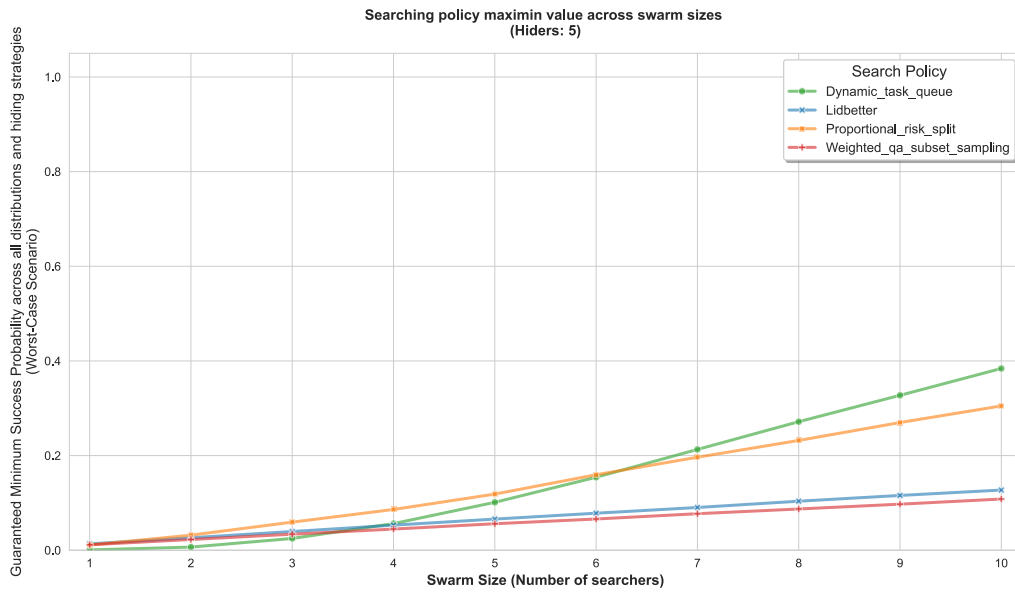


(c) Hiders: 3

Figure 5: Turnaround analysis (Part 1 of 2).



(d) Hiders: 4



(e) Hiders: 5

Figure 5: Turnaround analysis (Part 2 of 2).

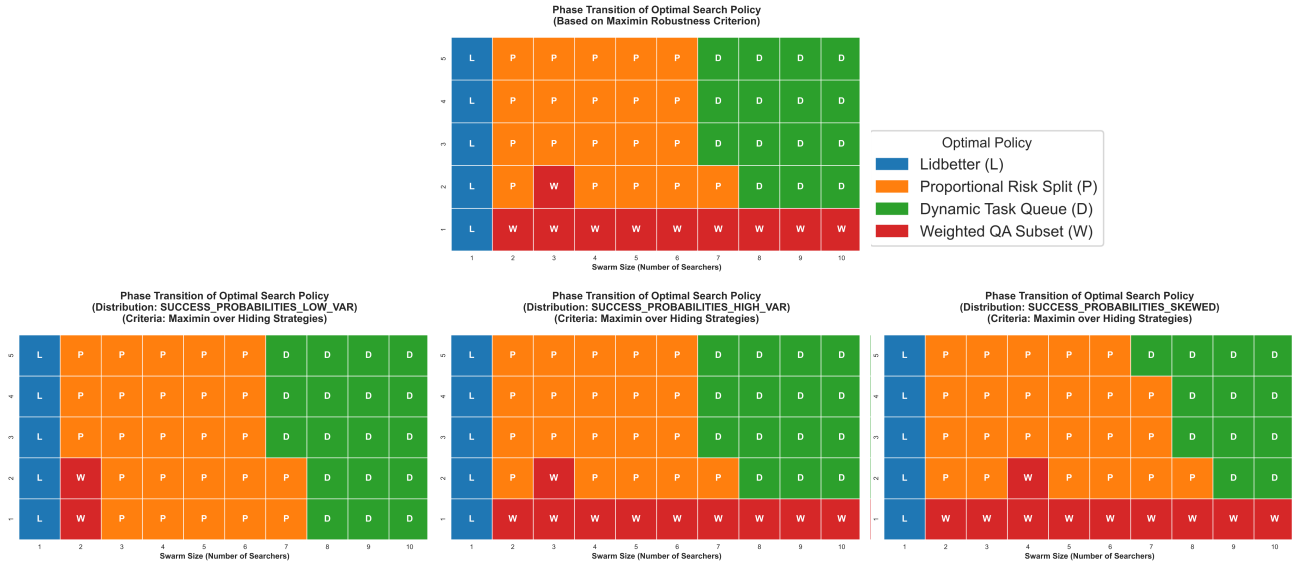


Figure 6: Transition matrix of the optimal search policy for a given swarm size (horizontal axis, 1 – 10), and a given number of hidrs (vertical axis, 1 – 5), regardless of hiding strategy. Top plot is regardless of probability distribution, bottom three from left to right is specified for the: Low variation, High variation, skewed probability distributions. Each individual marix can be found in the appendix ??,26,2728

6 Conclusion

The performance and robustness of various swarm search policies were evaluated against adversarial hiding strategies in a grid world environment. Through extensive Monte Carlo simulations across varying swarm sizes, hider counts, hiding strategies, and success probability distributions, distinct transitions in optimal policy selection driven by swarm size and hider density were identified.

In single-hider scenarios, where the hider ratio is 1/5, the optimal strategy for a single searcher is the game-theoretic optimum, specifically the Lidbetter searching policy. As the swarm size grows, the *Weighted qa subset sampling* policy consistently emerges as the most robust choice. This indicates that the game-theoretic optimum is scalable; *Weighted qa subset sampling* effectively applies the Lidbetter strategy individually across the drones in the swarm. While *Weighted qa subset sampling* should also be the optimum for a single drone (as this effectively is *Lidbetter*). The implementation specifics regarding the avoidance of non-target cells in the actual *Lidbetter* and not applying this to the *Weighted qa subset sampling* makes the latter more vulnerable to unnecessary risk and thus less effective.

In scenarios involving two hidrs (hider ratio 2/5), the optimal strategies shift away from *Weighted qa subset sampling*. While the Lidbetter policy remains the most robust for a single searcher, it ceases to be optimal for swarms. For a medium-sized swarm of 5 agents, the Proportional Risk Split proves to be the safest policy, whereas for larger swarms of 10 agents, the Dynamic Task Queue dominates. This latter policy achieved both the highest worst-case (π_{maximin}^*) and best-case (π_{max}) performance across all distributions.

Similar trends were observed in the saturated five-hider scenario (ratio 5/5), where hiding strategies became redundant. Although no clear single-searcher policy dominated—likely due to implementation artifacts rather than theoretical limitations—the swarm performance transitioned consistently: Proportional Risk Split was optimal for 5 drones, and Dynamic Task Queue was optimal for 10 drones. Notably, the specific success probability distribution of the environment did not significantly alter the identification of the safest searching policy.

Synthesizing these results reveals a "turnaround" effect as swarm size grows, indicated by a predictable evolution in the optimal searching policy. For individual searchers ($N = 1$), the Lidbetter strategy represents the theoretical optimum. For small to mid-sized swarms ($N \approx 2 - 6$), the Proportional Risk Split becomes the most robust approach by effectively balancing risk-based allocation with static routing. Finally, for large swarms ($N \geq 7$), the Dynamic Task Queue becomes the dominant policy, as the benefits of active communication and dynamic reallocation outweigh the efficiency of static pre-planning. Ultimately, there is no single "one-size-fits-all" policy. Optimal search performance requires selecting a strategy based primarily on swarm size and

secondarily on the expected number of targets, while the specific environmental probability distribution appears to be a less critical factor.

7 Discussion

The results of the analysis highlight distinct transitions in optimal policy selection based on swarm size. However, the analysis also revealed specific limitations in the current algorithm designs and identified promising avenues for future research to enhance robustness and performance.

7.1 Fusion of Dynamic Allocation and Risk Awareness

A primary recommendation for future work is the fusion of the *Dynamic Task Queue* with game-theoretic, risk-aware probabilistic sampling. In the current implementation, the *Dynamic Task Queue* assigns targets largely random or based on proximity (Manhattan distance) without prioritizing safer or higher-value targets first. While this proved effective for large swarms where volume compensates for risk, it lacks efficiency for smaller swarms. By integrating the risk-based prioritization of the *Proportional Risk Split* into the dynamic assignment logic, the policy could potentially become the most robust solution across all swarm sizes, rather than just large ones.

7.2 (Multi-Agent) Reinforcement Learning

While the report focused on heuristic and game-theoretic approaches, Multi-Agent Reinforcement Learning (MARL) presents a non-trivial alternative that should be investigated. RL agents have the potential to discover emergent coordination patterns that balance risk and coverage more effectively than static rule-based systems, potentially adapting dynamically to the specific probability distributions of the environment.

7.3 Non-Target Avoidance

A performance issue was found between the *Lidbetter* strategy and *Weighted qa subset sampling* for single-agent scenarios, despite the fact that they are theoretically similar. The *Weighted qa subset sampling* was frequently outperformed because it prioritized the shortest route to the target subset, whereas the *Lidbetter* implementation explicitly included logic for "non-target avoidance" (avoiding potential hiding locations when this is not the target to avoid unnecessary risk). The non-target avoidance definitely has benefits and should therefore be implemented in all strategies that allow for this.

7.4 Risk Weighted Tour

The *Risk Weighted Tour* performed poorly relative to other policies. A design flaw was identified in the cost function, which currently discounts distance based solely on the success probability of the *targeted* cell (P_{p2}), rather than accounting for the total survival probability of the entire route leading to that cell. To improve this strategy, the cost function should be updated to consider the total risk of the trajectory. Additionally, introducing a constant or dynamic factor to the discount mechanism could enhance the algorithm's ability to weigh safety against distance more effectively.

8 References

References

- Alfeo, A. L., Cimino, M. G., De Francesco, N., Lega, M., & Vaglini, G. (2018). Design and simulation of the emergent behavior of small drones swarming for distributed target localization. *Journal of Computational Science*, 29, 19-33. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1877750318302898> doi: <https://doi.org/10.1016/j.jocs.2018.09.014>
- Arnold, R., Mezzacappa, E., Jablonski, J., & Abruzzo, B. (2020). Multi-role uav swarm behaviors for wide area search using emergent intelligence. In *2020 fourth world conference on smart trends in systems, security and sustainability (worlds4)* (p. 255-261). doi: 10.1109/WorldS450073.2020.9210392
- Bram, J. (1963). *A 2-player n-region search game* (Tech. Rep.).
- Hoang, M.-T. O., Grøntved, K. A. R., van Berkel, N., Skov, M. B., Christensen, A. L., & Merritt, T. (2023). Drone swarms to support search and rescue operations: Opportunities and challenges. In B. J. Dunstan, J. T. K. V. Koh, D. Turnbull Tillman, & S. A. Brown (Eds.), *Cultural robotics: Social robots and their emergent cultural ecologies* (pp. 163-176). Cham: Springer International Publishing. Retrieved from https://doi.org/10.1007/978-3-031-28138-9_11 doi: 10.1007/978-3-031-28138-9_11
- Lidbetter, T. (2020, 2). Search and rescue in the face of uncertain threats. *European Journal of Operational Research*, 285(3), 1153-1160. Retrieved from <https://doi.org/10.1016/j.ejor.2020.02.029> doi: 10.1016/j.ejor.2020.02.029
- Lidbetter, T. (2025). A review of minimum cost box searching games. *arXiv preprint arXiv:2502.10551*.
- Nielsen, C. W., Few, D. A., & Athey, D. S. (2008). Using mixed-initiative human-robot interaction to bound performance in a search task. In *2008 international conference on intelligent sensors, sensor networks and information processing* (pp. 195-200).
- Rothfus, S., Chiou, M., Inga, J., Hohmann, S., & Stolkin, R. (2022, 10). A Negotiation-Theoretic Framework for control Authority transfer in Mixed-Initiative Robotic Systems. *2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 921-928. Retrieved from <https://doi.org/10.1109/smc53654.2022.9945196> doi: 10.1109/smc53654.2022.9945196

9 Appendix

9.1 Appendix - results

9.1.1 One hider

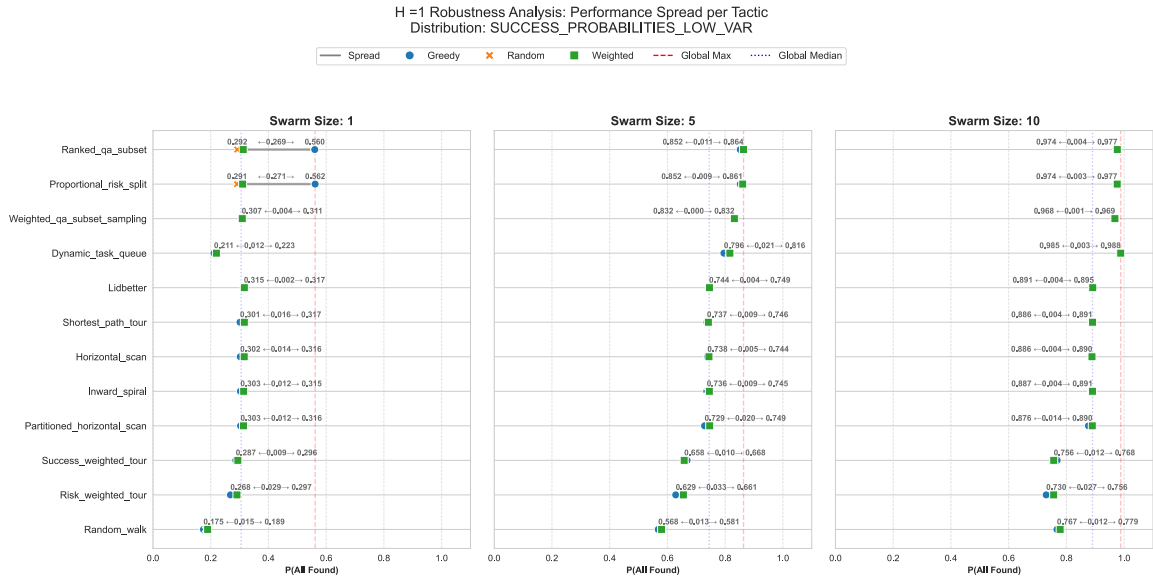


Figure 7: Cleveland dot plot of probability of all hiders found for swarm size 1, 5 and 10, given 1 hiders, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the low variation probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

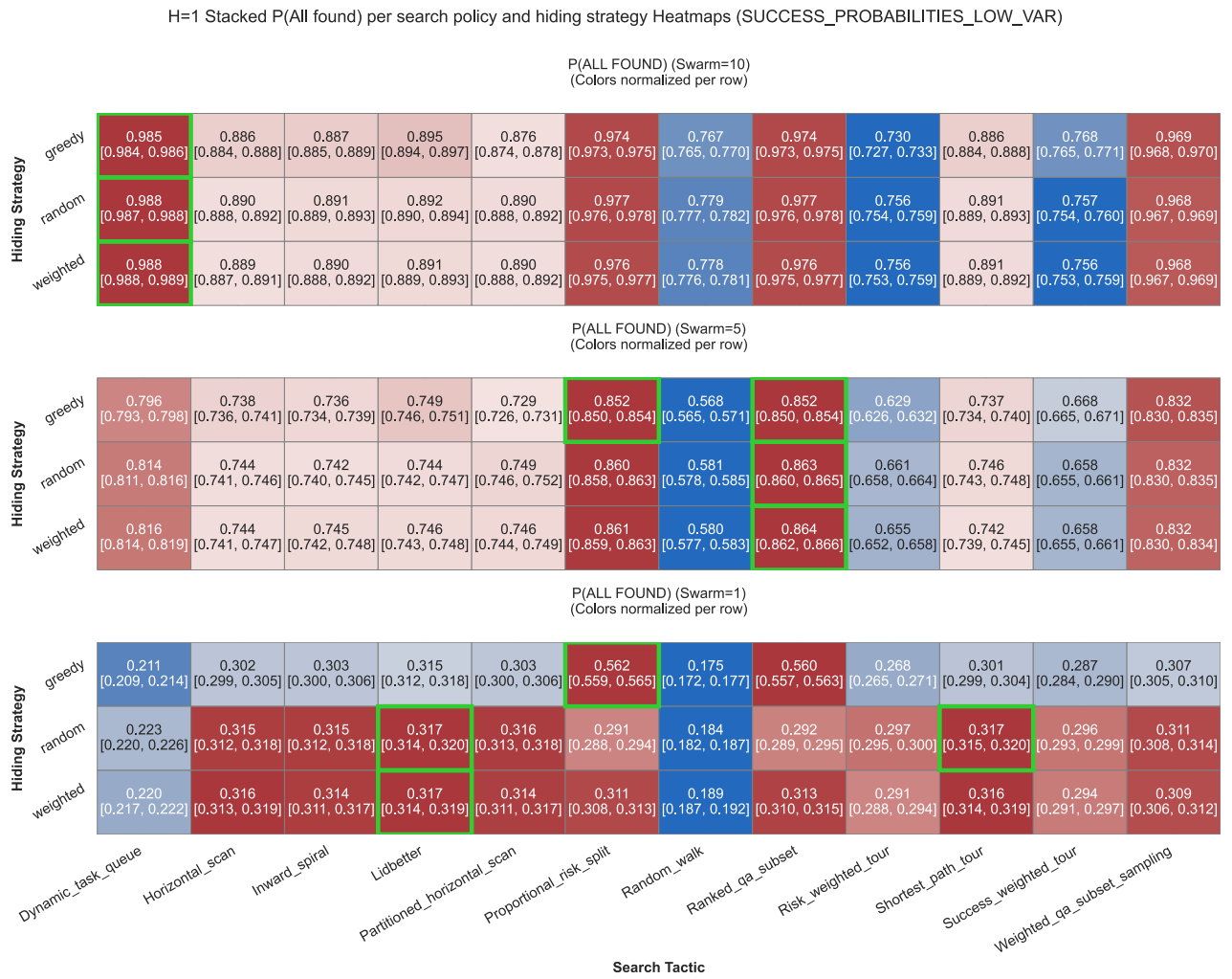


Figure 8: Full stacked heatmap of 1/5 hidens, all tactics, all hiding strategies, for the low variance probability distribution.

Low var

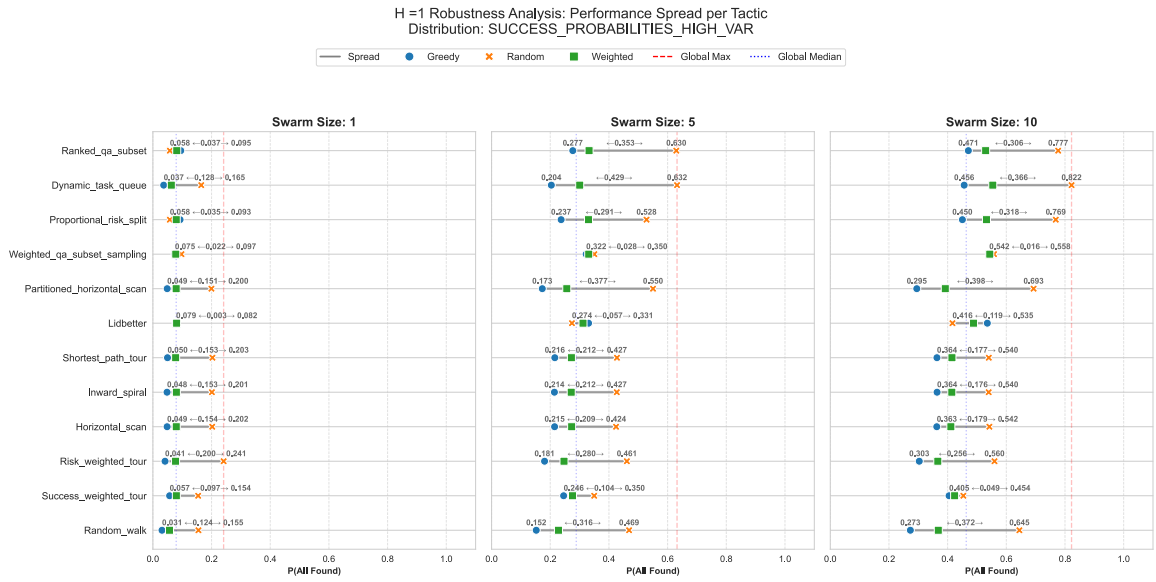


Figure 9: Cleveland dot plot of probability of all hidiers found for swarm size 1, 5 and 10, given 1 hidiers, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the high variation probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

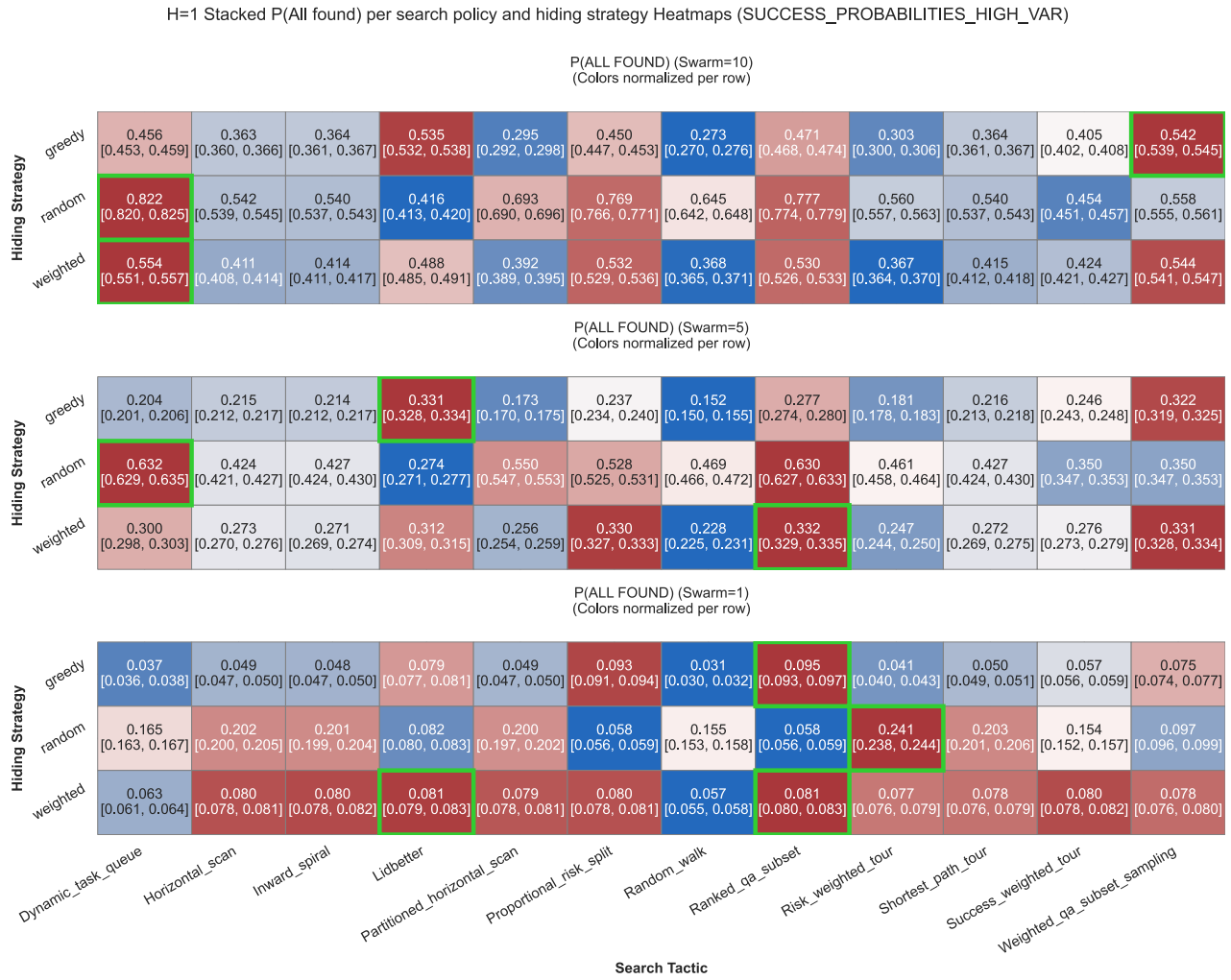


Figure 10: Full stacked heatmap of 1/5 hidiers, all tactics, all hiding strategies, for the HIGH variance probability distribution.

High var

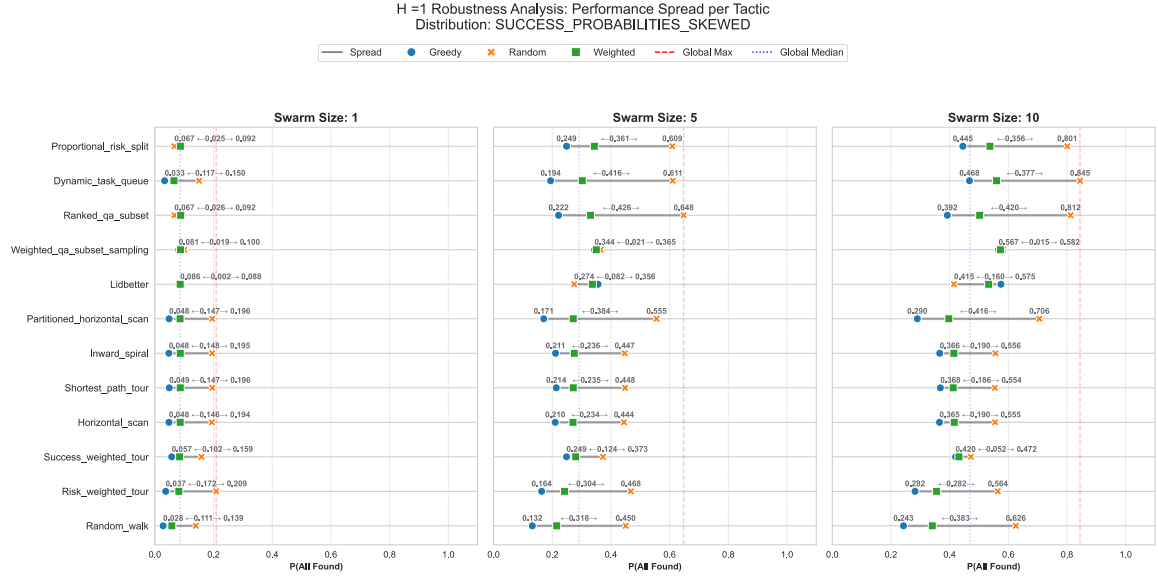


Figure 11: Cleveland dot plot of probability of all hidiers found for swarm size 1, 5 and 10, given 2 hidiers, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the skewed probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

H=1 Stacked P(All found) per search policy and hiding strategy Heatmaps (SUCCESS_PROBABILITIES_SKEWED)

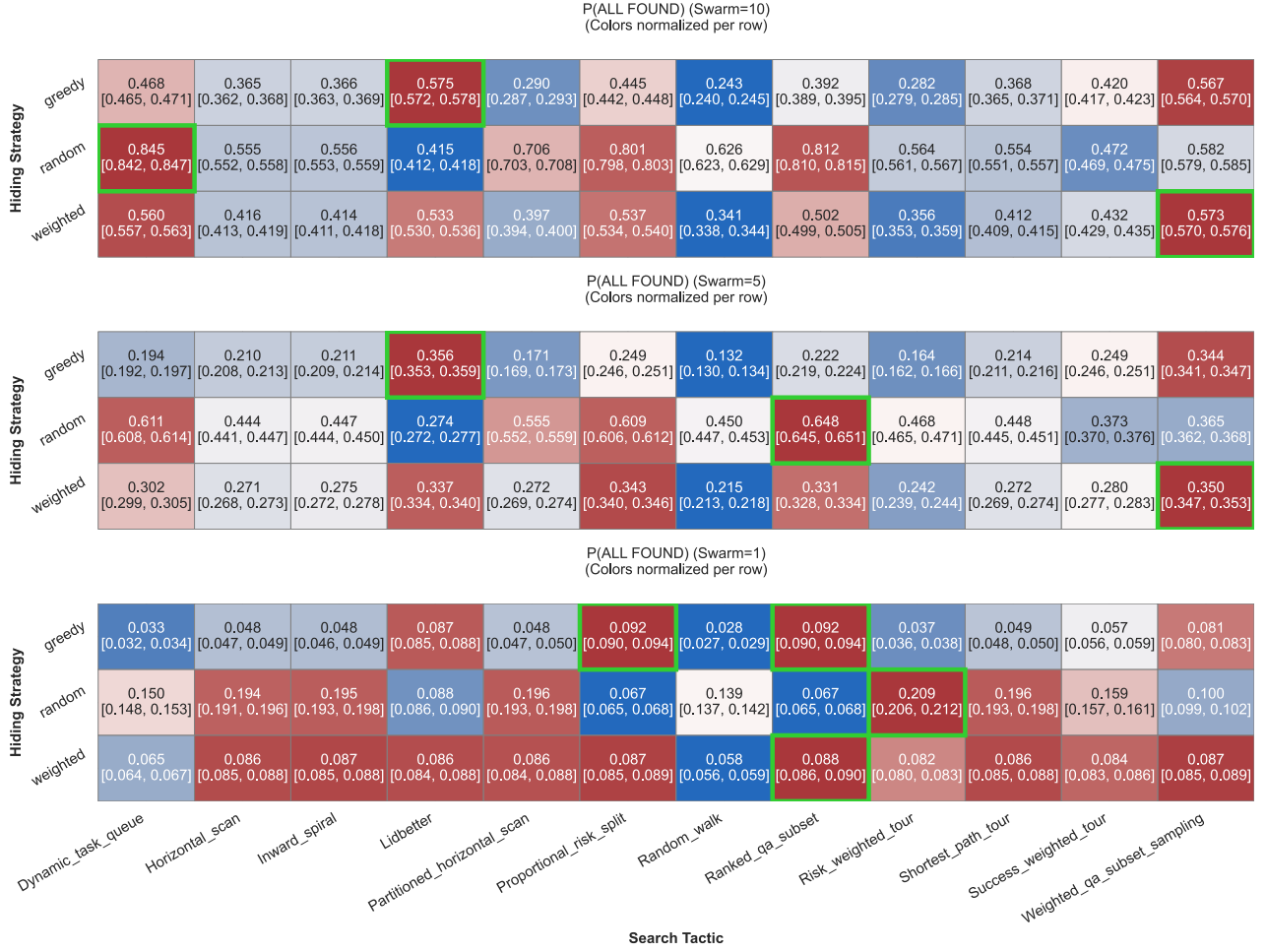


Figure 12: Full stacked heatmap of 1/5 hidere, all tactics, all hiding strategies, for the SKEWED probability distribution.

Skewed

9.1.2 Two hidrs

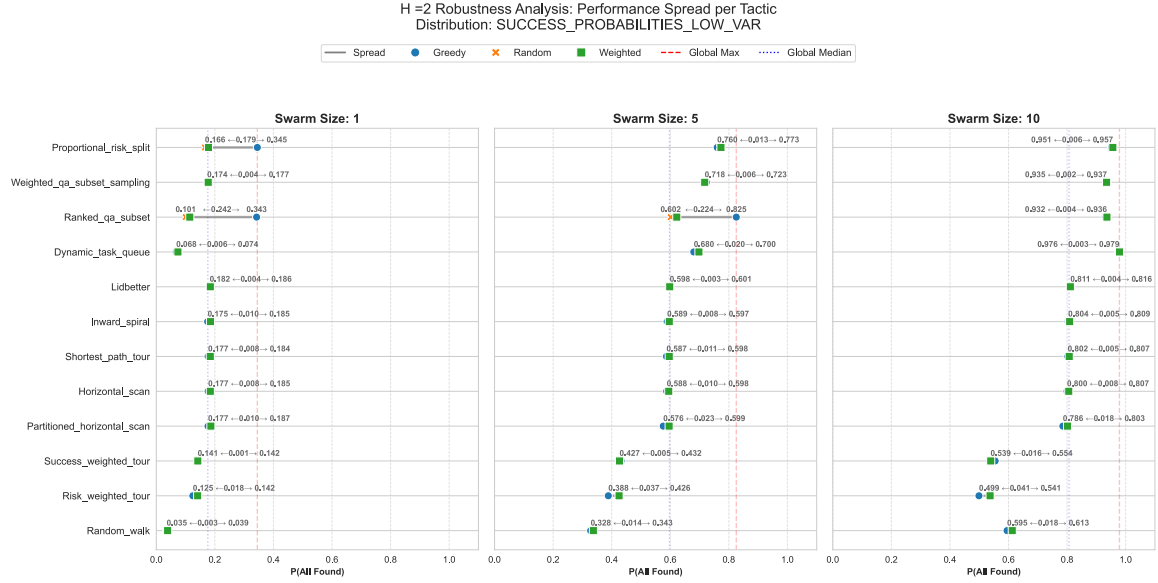


Figure 13: Cleveland dot plot of probability of all hidrs found for swarm size 1, 5 and 10, given 2 hidrs, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the low variation probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

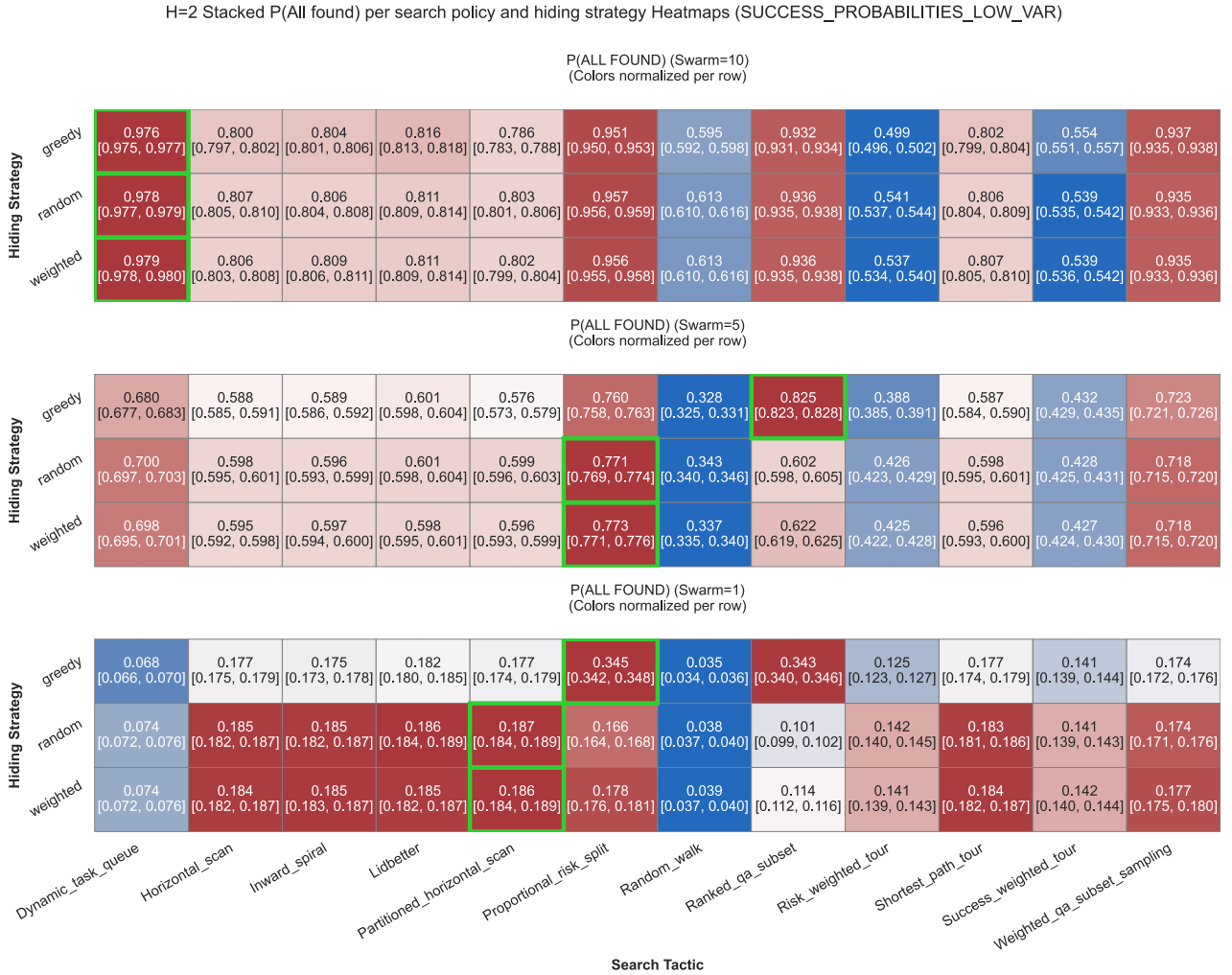


Figure 14: Full stacked heatmap of 2/5 hidrs, all tactics, all hiding strategies, for the low variance probability distribution.

Low var

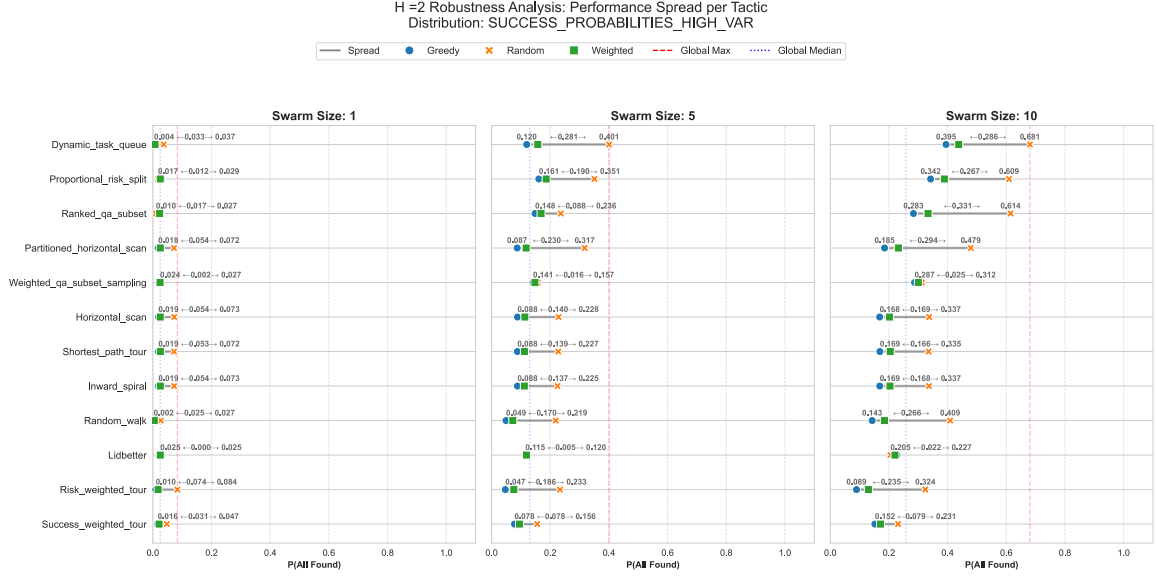


Figure 15: Cleveland dot plot of probability of all hidens found for swarm size 1, 5 and 10, given 2 hiders, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the high variation probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

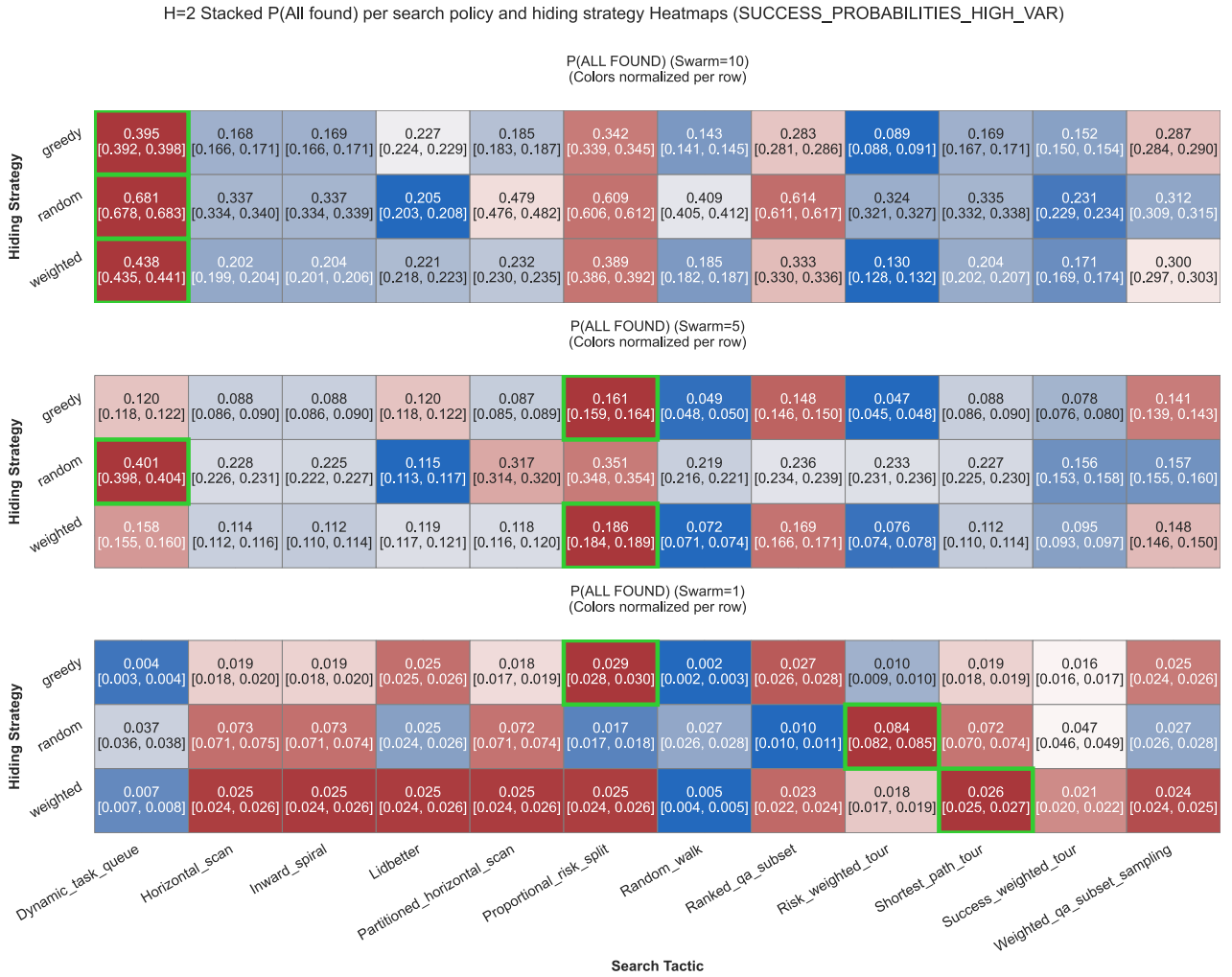


Figure 16: Full stacked heatmap of 2/5 hiders, all tactics, all hiding strategies, for the HIGH variance probability distribution.

High var

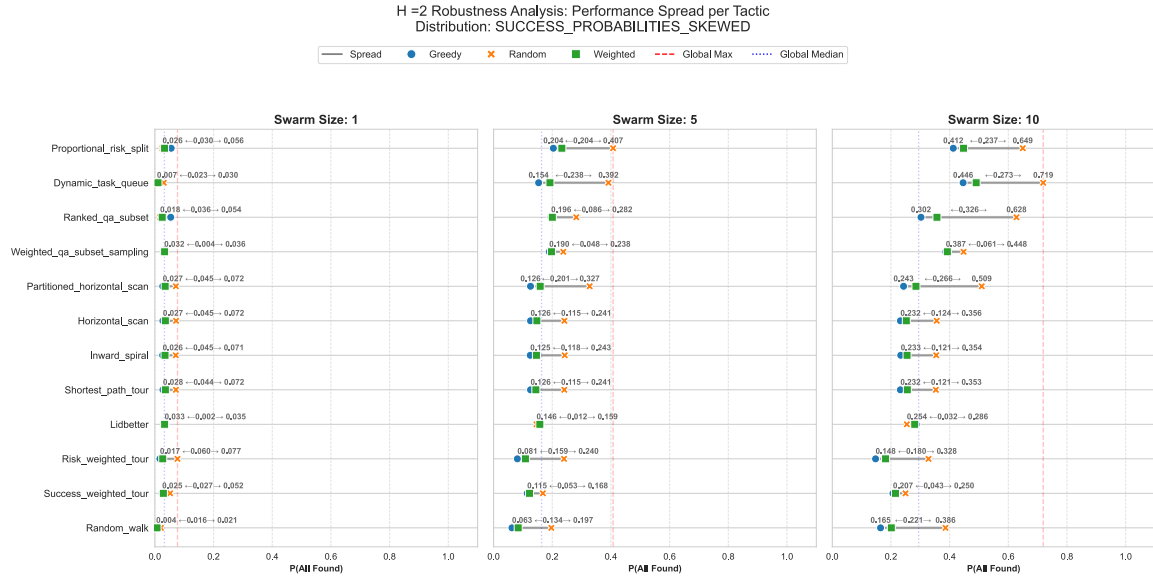


Figure 17: Cleveland dot plot of probability of all hidiers found for swarm size 1, 5 and 10, given 2 hidiers, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the skewed probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

H=2 Stacked P(All found) per search policy and hiding strategy Heatmaps (SUCCESS_PROBABILITIES_SKEWED)

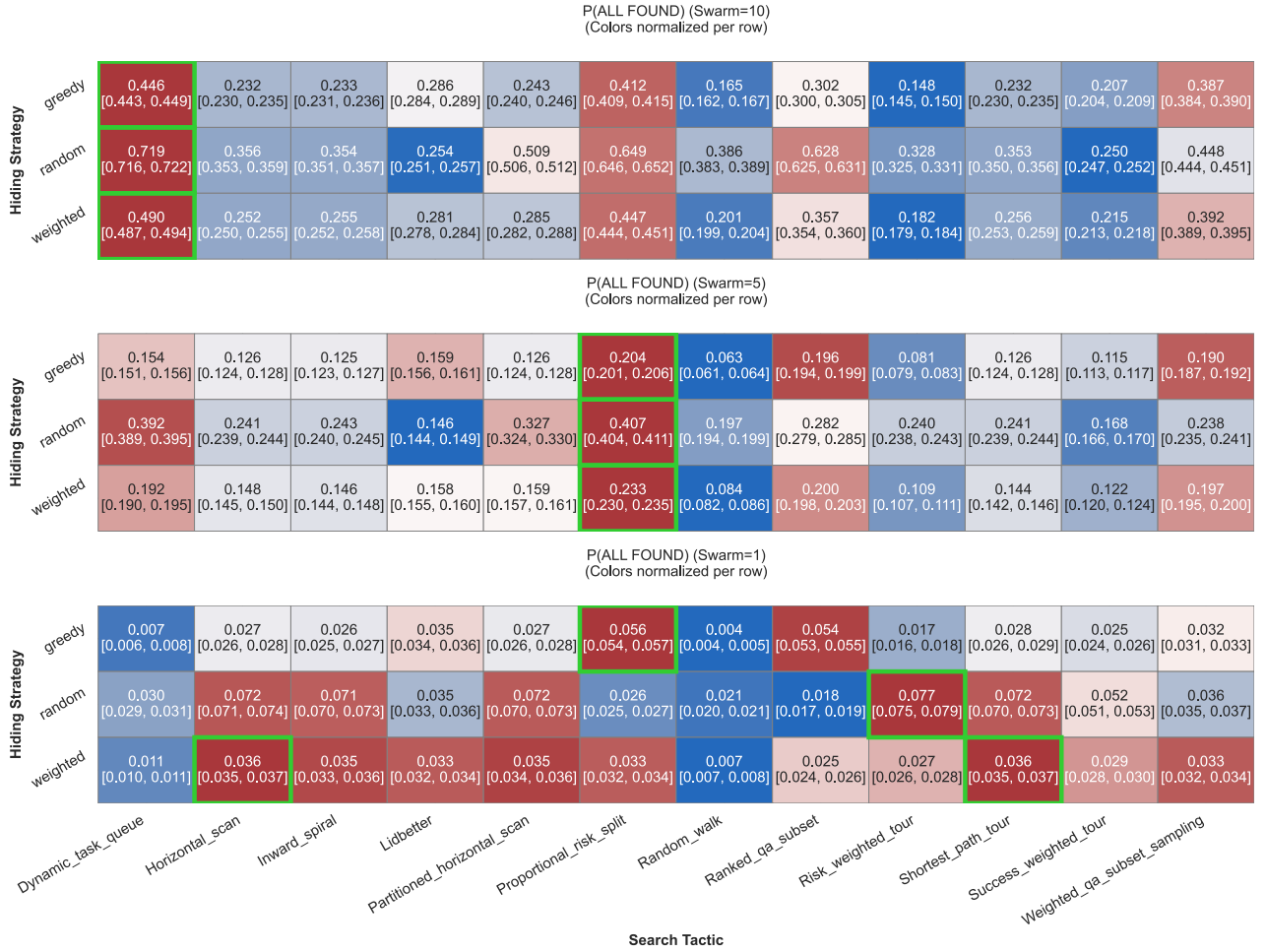


Figure 18: Full stacked heatmap of 2/5 hidiers, all tactics, all hiding strategies, for the SKEWED probability distribution.

Skewed

9.1.3 Five hidrs

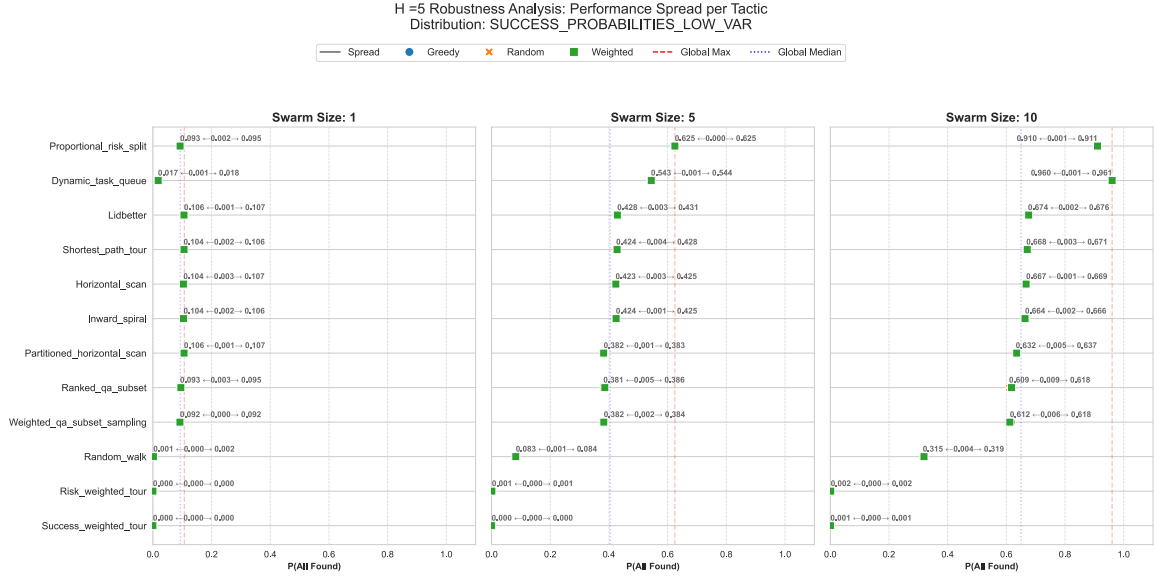


Figure 19: Cleveland dot plot of probability of all hidrs found for swarm size 1, 5 and 10, given 5 hidrs, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the skewed probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

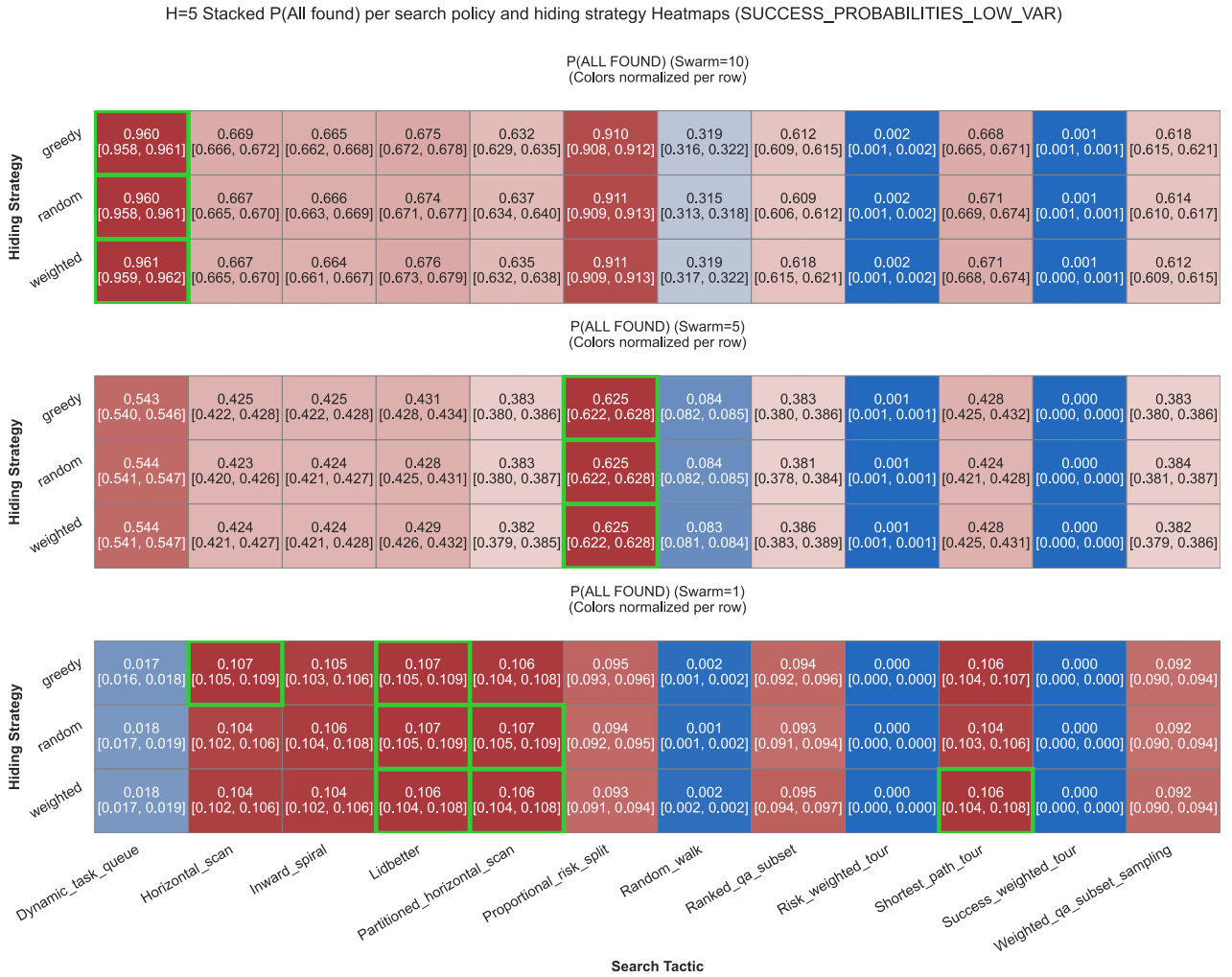


Figure 20: Full stacked heatmap of 5/5 hidrs, all tactics, all hiding strategies, for the low variance probability distribution.

Low var

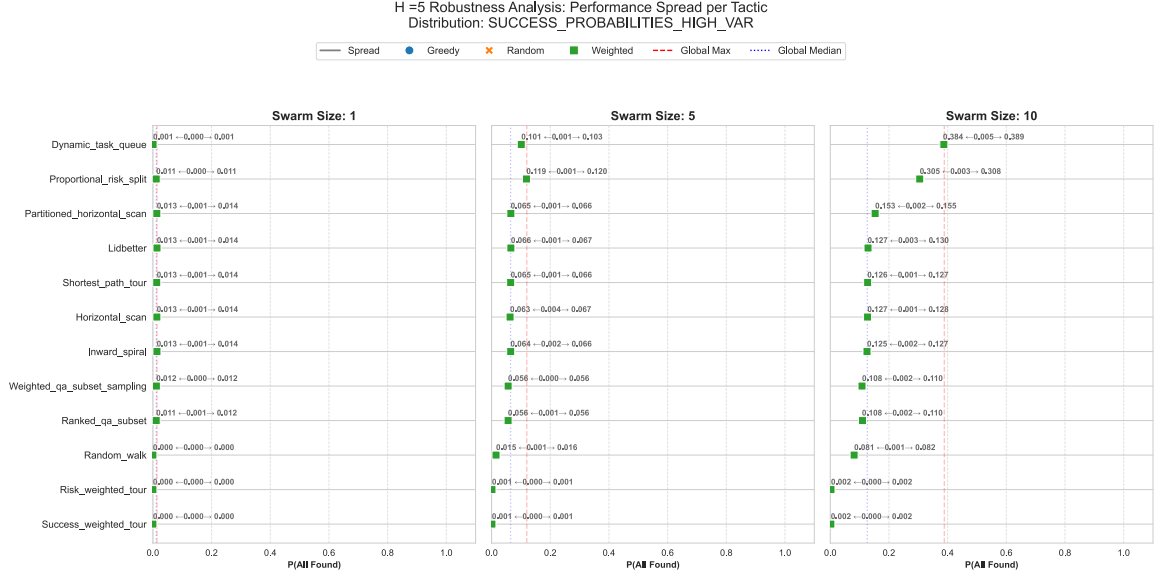


Figure 21: Cleveland dot plot of probability of all hiders found for swarm size 1, 5 and 10, given 5 hiders, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the skewed probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

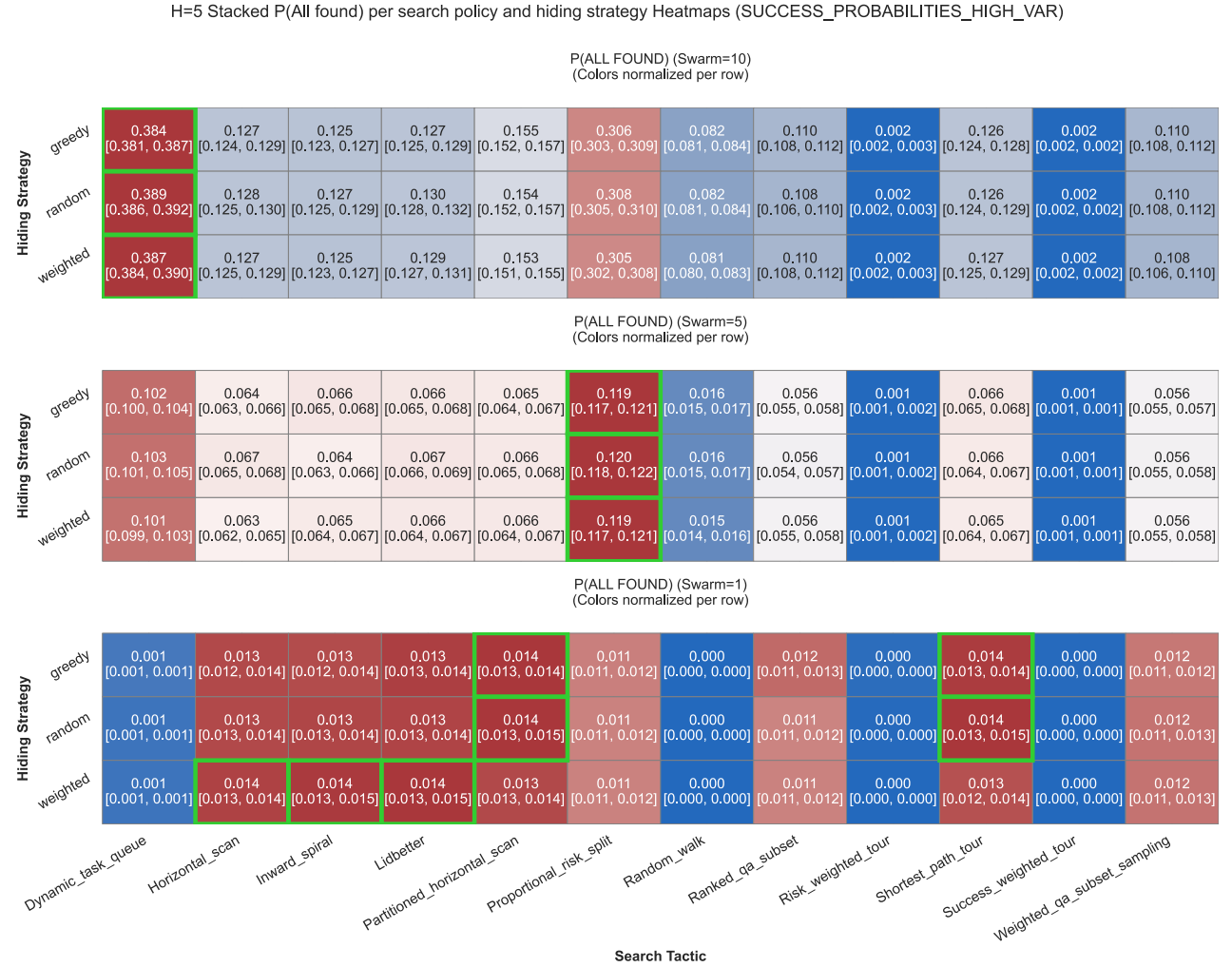


Figure 22: Full stacked heatmap of 5/5 hiders, all tactics, all hiding strategies, for the HIGH variance probability distribution.

High var

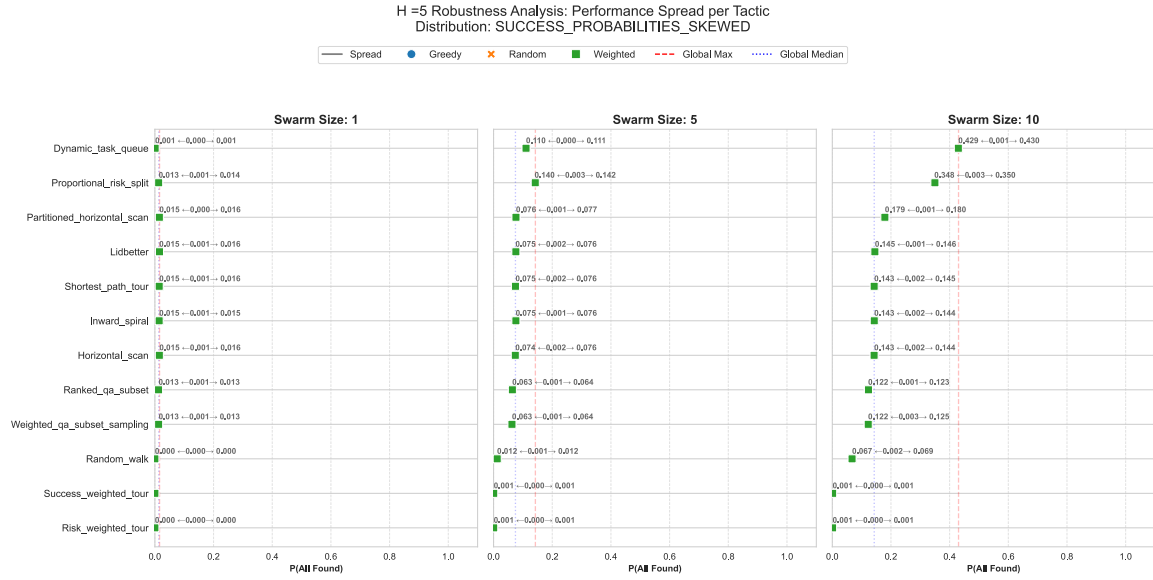


Figure 23: Cleveland dot plot of probability of all hidiers found for swarm size 1, 5 and 10, given 5 hidiers, 5 possible hiding locations and the different hiding strategies, simulated on a 20 x 20 grid with the skewed probability distribution. The plot shows robustness of strategies, i.e. wider lines could be viewed as less robust.

H=5 Stacked P(All found) per search policy and hiding strategy Heatmaps (SUCCESS_PROBABILITIES_SKEWED)

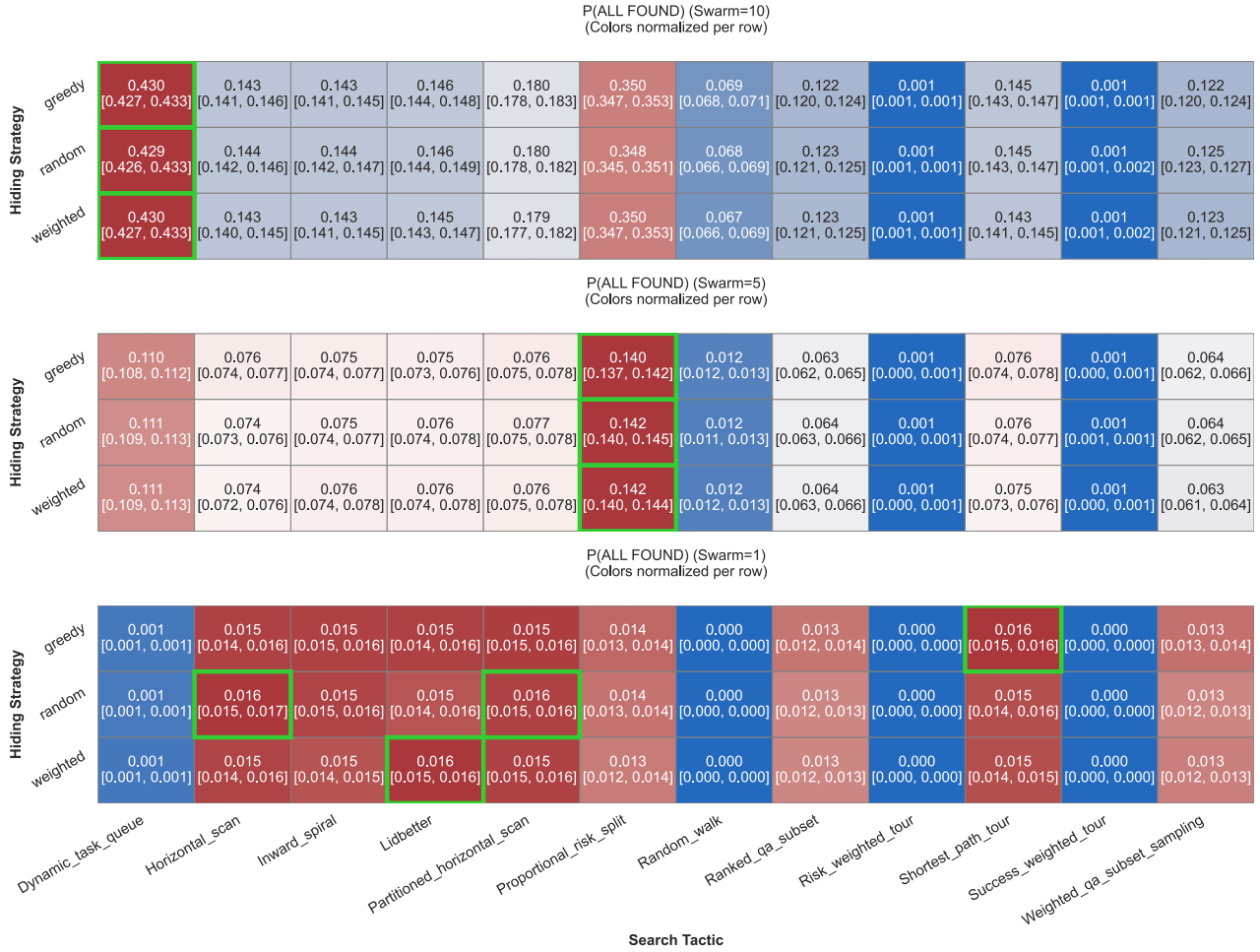


Figure 24: Full stacked heatmap of 5/5 hidere, all tactics, all hiding strategies, for the SKEWED probability distribution.

Skewed

9.2 Transition / decision matrices

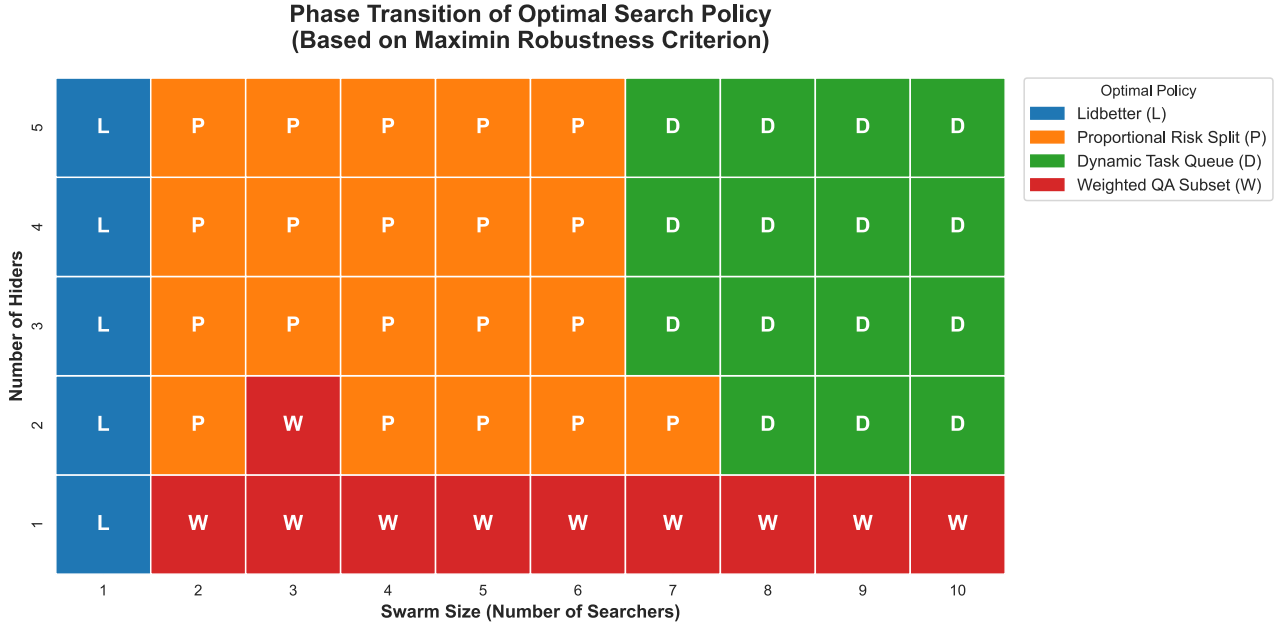


Figure 25: Phase transition / decision matrix. Shows what is the most robust strategy for each hider $\times$ swarm size combination based on the π_{maximin}^* .

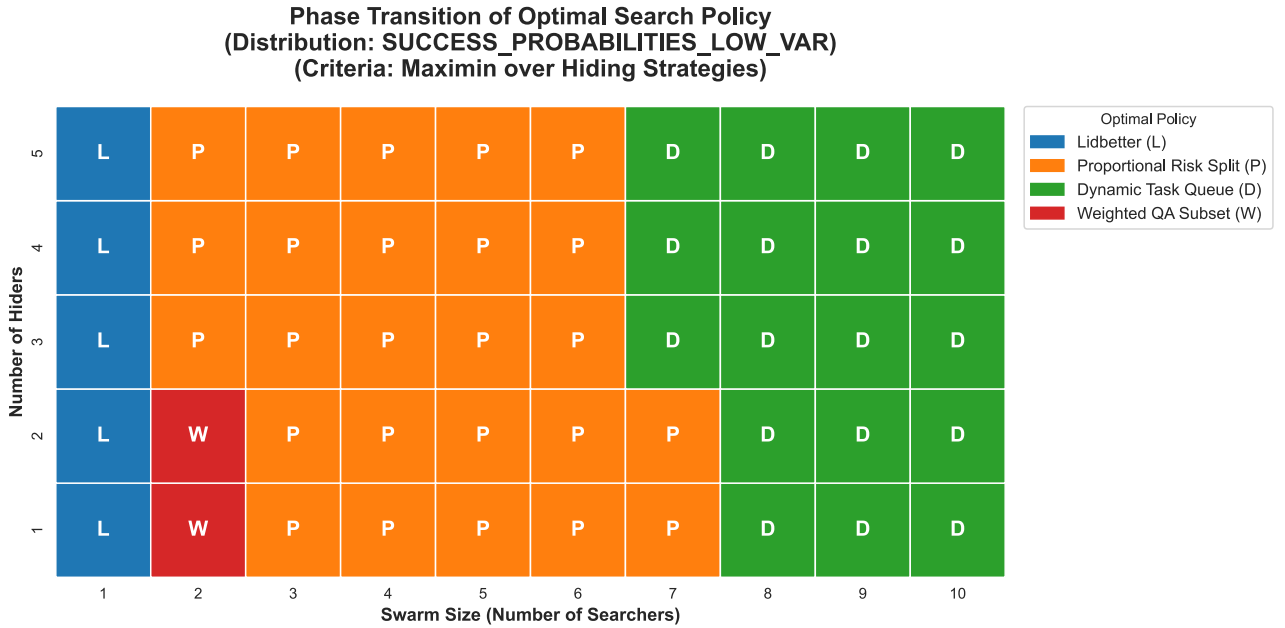


Figure 26: Phase transition / decision matrix. Shows what is the most robust strategy for each hider $\times$ swarm size combination based on the π_{maximin}^* , specific for low variance probability distribution.

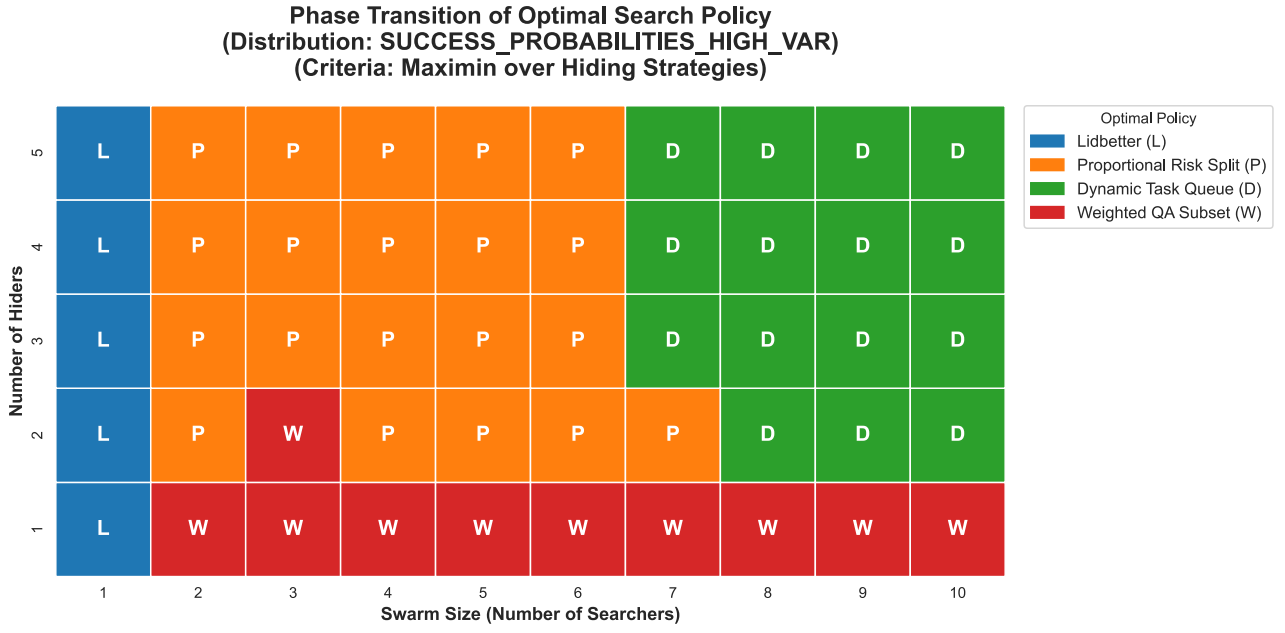


Figure 27: Phase transition / decision matrix. Shows what is the most robust strategy for each hider $\times$ swarm size combination based on the π_{maximin}^* , specific for high variance probability distribution.

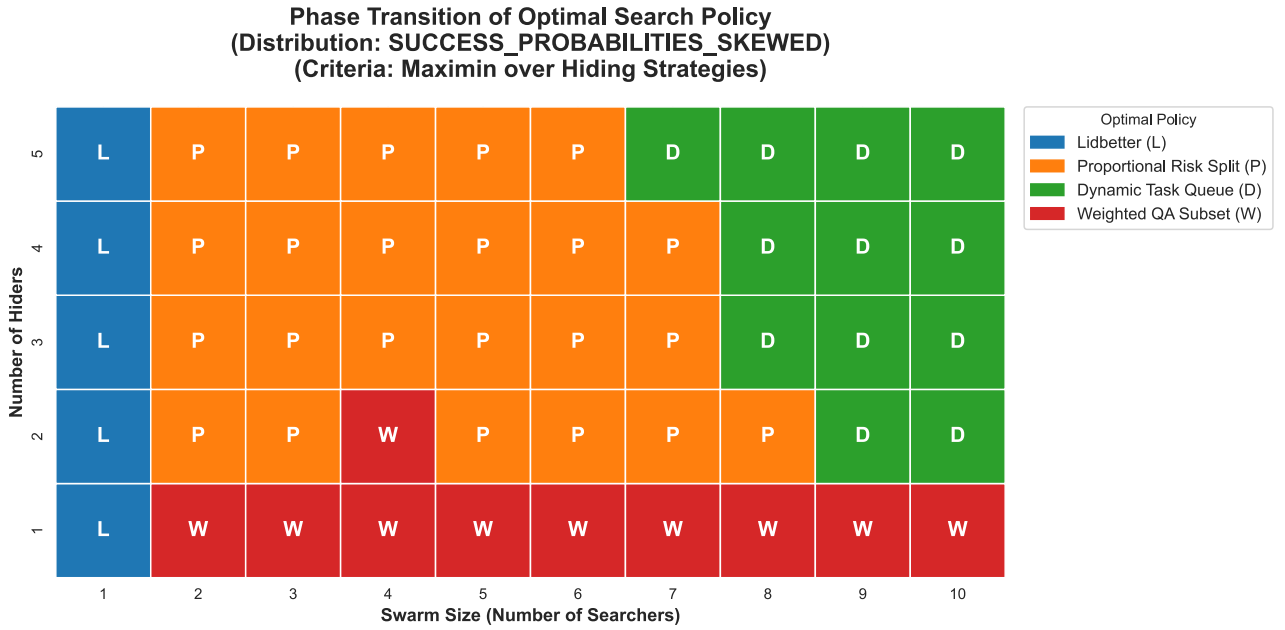


Figure 28: Phase transition / decision matrix. Shows what is the most robust strategy for each hider $\times$ swarm size combination based on the π_{maximin}^* , specific for skewed probability distribution.